

**Predicting SME Export Status in the East African Community: A
Machine Learning Approach Benchmarked Against a European
Firm-Level Model**

CHAPTER 1. INTRODUCTION

1.1 Background and Context

International trade is a principal driver of economic growth, productivity and long-term development. Small and medium-sized enterprises (SMEs) are widely treated as a development lever in low- and middle-income economies: they account for the majority of registered firms and a large share of employment, and export activity is repeatedly identified as one of the pathways through which they raise productivity, adopt new technology, and grow (International Trade Centre, 2025). In Southern Africa, firm-level evidence from the World Bank Enterprise Surveys (WBES) shows that exporting status is itself a significant predictor of subsequent firm growth, alongside firm size, age, and technology use (Masenyetse and Manamathela, 2023). Within East Africa specifically, the Customs Union established under the East African Community (EAC) in 2005 was designed to lower intra-regional trade barriers and give exporting SMEs a wider market to grow into (East African Community, n.d.); yet recent research finds that intra-EAC trade continues to fall well short of the bloc's integration potential, with infrastructure and non-tariff barriers still constraining firms that might otherwise export within the region (Hersi, Alasow and Omar, 2025). Similarly, the World Bank estimates that formal SMEs contribute up to 40% of GDP in emerging economies and create roughly seven of every ten formal jobs, underscoring their importance for inclusive, sustainable growth (World Bank, 2020).

Understanding what distinguishes exporting SMEs from non-exporting SMEs has become an important research and policy priority. Prior empirical studies identify numerous determinants of export participation, including firm size, productivity, innovation capability, access to finance, ownership structure, workforce skills, managerial experience and technology adoption (Love & Roper, 2015; Haddoud et al., 2021). Most rely on conventional econometric techniques such as logistic regression, probit and linear regression to estimate the probability of export participation. While these methods have generated valuable theoretical insight into the drivers of internationalisation, they generally impose linearity and predefined functional forms, which may limit their ability to capture the complex, nonlinear interactions that often characterise firm behaviour in dynamic, heterogeneous business environments.

Alongside this explanatory literature, a related but largely distinct strand approaches exporting from a predictive perspective. Rather than asking why firms export, this emerging work asks

whether routinely collected firm-level data can identify which non-exporting firms are most likely to become future exporters. This "exporter-prediction problem" shifts the focus from explanation to decision support, developing models that help governments, export promotion agencies, financial institutions and development organisations allocate scarce resources toward firms with the greatest export potential (Micocci & Rungi, 2023). Exporter prediction thus bridges academic research and evidence-based policymaking by turning firm-level data into actionable insight for export promotion.

Recent advances in artificial intelligence (AI) and machine learning (ML) address many of the methodological limitations of traditional econometric approaches. Unlike conventional statistical models, ML algorithms are designed to maximise predictive accuracy and can identify complex nonlinear relationships and hidden patterns in large, high-dimensional datasets without imposing restrictive assumptions. Machine learning has consequently been adopted across finance, healthcare, marketing and economics to support prediction, classification and decision-making (James et al., 2021). Within international business research, growing evidence suggests ML techniques hold considerable potential for predicting firms' export behaviour: Micocci and Rungi (2023) demonstrated that ML algorithms can successfully identify firms with high export potential, providing practical decision-support tools capable of improving the efficiency of export promotion policy.

The growing availability of comprehensive firm-level datasets has further accelerated ML's application in business and economic research. Among the richest sources of enterprise-level information are the World Bank Enterprise Surveys (WBES), which collect harmonised data on firm characteristics, business performance, access to finance, innovation, infrastructure, ownership, workforce composition, technology adoption and export activity across developing economies. The breadth, consistency and cross-country comparability of these data provide an appropriate empirical foundation for predictive models capable of identifying SMEs with a high likelihood of exporting. The availability of harmonised survey data across multiple EAC member states, in particular, offers an opportunity to examine export participation regionally while accounting for differences in institutional environments and firm characteristics.

Recent developments in trade policy reinforce this research's relevance. The SME Competitiveness Outlook 2025, published by the International Trade Centre, highlights digital capability, data-

driven decision-making and information readiness as increasingly important determinants of SME competitiveness in global markets, and warns that firms in developing economies risk falling further behind if they lack the digital infrastructure and analytical capability required to compete internationally (International Trade Centre, 2025). Within this context, a validated, low-cost, survey-based exporter-prediction model offers a practical policy solution: unlike approaches requiring detailed financial statements or sophisticated digital accounting systems, predictive models based on WBES data need only information collected through professionally administered enterprise surveys, making them well suited to developing-country contexts where detailed financial data are often unavailable or incomplete (World Bank Group, n.d.).

Against this background, this dissertation applies supervised machine learning to predict export participation among SMEs in the East African Community using World Bank Enterprise Survey data. It builds on the exporter-prediction framework developed and validated on European firm-level data by Micocci and Rungi (2023) and examines whether that approach transfers successfully to a Sub-Saharan African context characterised by different institutional environments, business constraints and levels of economic development. By comparing the predictive performance of multiple machine learning algorithms and identifying the firm characteristics that contribute most to export participation, the study extends the literature on SME internationalisation while demonstrating a practical application of artificial intelligence in international business research. The findings are intended to provide policymakers, export promotion agencies, development organisations and financial institutions with evidence to design interventions that improve SME competitiveness, increase export participation and promote sustainable economic development across the EAC.

1.2 Problem Statement and Research Gap

The exporter-prediction problem has, to date, been demonstrated convincingly in only one empirical setting. Micocci and Rungi (2023) trained a set of machine learning classifiers logit, LASSO-regularised logit, classification trees, random forests, and a Bayesian Additive Regression Tree with Missingness in Attributes (BART-MIA) on ORBIS financial data for French firms observed between 2010 and 2018, finding that BART-MIA achieved accuracy of up to 0.90 in separating exporters from non-exporters, with the resulting "exporting scores" argued to be useful for trade promotion and trade-credit targeting. That model was built and validated on a single,

high-income, single-country dataset with rich audited financial statements. Whether a BART-MIA-based approach, or a simpler benchmark such as logit or random forest, retains anything like comparable performance when the underlying data are Sub-Saharan African, firm-survey-based rather than financial-statement-based, and pooled across three EAC economies rather than one, is not established in the literature. This is the gap the thesis addresses: not a new modelling technique, but a benchmark-replication test of an existing, well-validated technique in a context (Kenya, Uganda and Tanzania, using WBES data) that differs from the original along data type, income level, institutional setting and regional trade architecture. Establishing whether, and how well, the approach generalises is a precondition for using it as a scoring tool for EAC trade-promotion agencies.

1.3 Research Questions and Objectives

1. Which firm-level features are most predictive of export status among SMEs in the East African Community?

Objective: To identify and rank, using SHAP-based feature importance, the WBES firm characteristics most strongly associated with observed export status across the pooled EAC sample.

2. How does model performance on EAC SME data compare with the benchmark established by Micocci and Rungi (2023) on European firm data?

Objective: To replicate the Micocci and Rungi (2023) approach, including BART-MIA, on a matched-prevalence manufacturing subsample of EAC firms, and compare accuracy, AUC-ROC and related metrics directly against the published European benchmark.

3. Can a context-adapted version of the model deliver a practically useful accuracy gain over the direct European benchmark when applied to the full-sector EAC sample?

Objective: To evaluate all five candidate models (logit, LASSO-logit, CART, random forest, BART-MIA) on the full-sector EAC dataset and assess whether performance is sufficient to support a prototype scoring tool for trade-promotion use.

4. Are the predictors of export status consistent across the three EAC economies studied, or do they vary by country?

Objective: To compare SHAP feature-importance rankings separately for Kenya, Uganda and Tanzania and assess the degree of convergence or divergence in predictor importance across the three economies.

1.4 Significance and Contribution

The thesis makes two contributions. The first is practical: if a benchmarked model performs adequately on EAC data, the output is a candidate scoring tool that trade-promotion agencies in

Kenya, Uganda or Tanzania could use to prioritise SME outreach, in the spirit of Micocci and Rungi's (2023) proposal for trade promotion and trade-credit targeting in Europe. Given that the EAC Customs Union already frames intra-regional trade facilitation as a core objective (East African Community, n.d.), and that SME digital and export readiness is a stated global policy priority (International Trade Centre, 2025), a validated regional scoring approach would be directly actionable rather than purely academic.

The second contribution is methodological: this is a cross-context benchmark replication with explicit variable harmonisation, rather than a novel-model paper. Comparable exporter-prediction work in African settings has tended to rely on constructed readiness indices or small primary surveys rather than large, nationally representative, government-grade firm surveys; testing an established European benchmark against real observed export status in a representative EAC firm sample is therefore a distinct empirical contribution to the small literature connecting machine learning to SME internationalisation research in developing economies. It also responds directly to the finding that exporting is itself a significant driver of firm growth in African firm-survey data (Masenyetse and Manamathela, 2023): if export status can be predicted before it is observed, the same data can in principle target future growth-promoting interventions, not merely explain past growth.

1.5 Scope and Delimitations

The thesis is scoped in three deliberate ways, each justified fully in Chapter 3. First, the geographic and data scope is limited to three EAC economies, Kenya, Uganda and Tanzania, using World Bank Enterprise Survey rounds (Kenya and Uganda, 2025; Tanzania, 2023), pooled to 2,229 firms, of which 297 (13.3%) report exporting. Second, the analysis uses a dual sector-scope design: the full-sector pool is the primary, applied dataset, while a manufacturing-only subsample (827 firms, 22.4% exporters) serves as a matched-prevalence dataset for direct benchmark comparison against Micocci and Rungi (2023), since their French sample and this thesis's full-sector EAC sample differ substantially in sectoral composition and export prevalence. Third, and most importantly, this is a secondary-data-only study: no primary data collection was undertaken, and all analysis relies on the WBES's professionally implemented stratified sampling design (World Bank Group, n.d.). This scope necessarily excludes firm-level information that only primary fieldwork could

capture, and the pooled EAC sample remains an order of magnitude smaller than the French benchmark dataset, a constraint returned to directly in the limitations discussion in Chapter 5.

1.6 Structure of the Thesis

The thesis is organised into six chapters. Chapter 1 (this chapter) presents the background, importance of international trade, the role of SMEs in economic development, and the challenges of SME export participation within the EAC. Chapter 2 reviews the literature on SME internationalisation and export behaviour, firm-level determinants of export status, machine learning approaches to trade prediction, the Micocci and Rungi (2023) anchor study, related African export-readiness work, and constraints facing SME exporters in Sub-Saharan Africa and the EAC, closing with a synthesis table mapping each research question onto the relevant literature strand and gap. Chapter 3 explains the research philosophy, design and methodology, describing the WBES dataset, sample selection, variable definition, preprocessing, feature engineering, the supervised learning algorithms used, evaluation metrics, and model-interpretation techniques. Chapter 4 presents the empirical findings, beginning with descriptive statistics and exploratory analysis before reporting model performance benchmarked against Micocci and Rungi (2023), together with SHAP-based feature-importance results overall and by country. Chapter 5 interprets the findings in relation to the existing literature and theoretical framework, evaluates their theoretical and practical implications, situates them relative to the European benchmark, and states limitations and future-research directions. Chapter 6 concludes with a summary of findings, the thesis's contribution to knowledge, and recommendations for practice and policy.

CHAPTER 2. LITERATURE REVIEW

2.1 Introduction

This chapter reviews the theoretical, empirical and methodological literature relevant to predicting SME export participation. It establishes the theoretical foundations of firm export behaviour, identifies firm-level characteristics associated with export participation, evaluates methods previously used to predict exporter status, and establishes the research gap addressed by this study. Particular attention is given to developing economies and the East African Community (EAC), where SMEs operate under institutional, financial and infrastructural conditions that differ substantially from the developed economies in which most exporter literature has been produced.

Export participation matters because internationalisation gives firms access to larger markets, new technologies and productivity growth. Firms do not participate uniformly, however: exporters tend to differ systematically from non-exporters in productivity, size, financial capacity and technological capability (Melitz, 2003; Bernard et al., 2007), differences that matter particularly for SMEs, which generally have fewer resources to absorb the fixed costs and risks of foreign-market entry.

Most empirical SME export studies take an explanatory approach, using logistic regression, probit and panel regression to identify statistically significant determinants of exporting. These generate valuable theoretical insight, but prediction is a different objective: rather than estimating the average effect of individual variables, predictive modelling asks whether observable firm characteristics can accurately distinguish exporters from non-exporters on unseen data. Machine learning offers new capacity to capture nonlinear relationships without pre-specifying them. Micocci and Rungi (2023), for example, show exporter status can be predicted from firm-level information using machine learning, with flexible ensemble methods outperforming conventional linear approaches in their European application; comparable evidence for African SMEs, particularly within the EAC, remains limited.

This chapter therefore integrates three strands of literature theories of why firms differ in export propensity, empirical evidence on firm-level determinants, and methodological research on machine learning and explainable artificial intelligence (XAI) before examining the anchor study

of Micocci and Rungi (2023), reviewing evidence specific to Sub-Saharan Africa and the EAC, and closing with a synthesis of the research gap and conceptual framework.

2.2 Theoretical Foundations of SME Export Behaviour

This study's theoretical foundation derives principally from the Resource-Based View (RBV) and heterogeneous-firm trade theory, complemented by the self-selection and learning-by-exporting literature.

2.2.1 Resource-Based View

RBV argues that differences in firm performance arise partly from differences in the resources and capabilities firms control. Barney (1991) proposes that resources generating sustained competitive advantage are valuable, rare, imperfectly imitable and non-substitutable including tangible assets such as financial capital and productive capacity, and intangible capabilities such as managerial expertise, technological competence and innovation capability. Applied to SME internationalisation, RBV suggests firms with stronger resource bundles are better positioned to absorb the costs and uncertainties of foreign-market entry, since exporting often requires investment in market research, certification, product adaptation, logistics, technology and compliance with foreign standards. This is especially relevant here because such resources are observable, directly or via proxies, in surveys like the WBES: firm size, access to finance, quality certification, ownership, workforce characteristics and technological activity can be read as manifestations of the resource bundle underlying export capability. RBV nevertheless has limits its emphasis on internal resources may understate external institutional conditions such as financial-system development, infrastructure and regional integration, which matter particularly in developing economies where institutional constraints can prevent capable firms from converting resources into actual exports. RBV is therefore used here as one theoretical foundation rather than a complete explanation.

2.2.2 Heterogeneous-Firm Theory and Self-Selection

A complementary foundation is Melitz's (2003) heterogeneous-firm model, which shows that firms differ in productivity and that this determines their ability to absorb the fixed costs of entering export markets. Only sufficiently productive firms can profitably internationalise; less productive firms remain domestic, and the least productive may exit under increased international

competition. Exporters and non-exporters are therefore expected to differ systematically rather than being drawn randomly from the same population, and Bernard et al. (2007) confirm empirically that exporters tend to be larger, more productive and more capital intensive. For this dissertation, Melitz's significance extends beyond productivity itself: it justifies treating exporter status as predictable from firm-level characteristics, since a classification model should be able to learn part of the distinction if exporters systematically differ from non-exporters. The relationship to Melitz (2003) is thus one of theoretical grounding rather than direct replication the present study operationalises the broader proposition using observable WBES characteristics within a predictive framework.

2.2.3 Self-Selection and Learning-by-Exporting

The relationship between exporting and firm performance is complicated by the learning-by-exporting hypothesis: while self-selection holds that better-performing firms become exporters, learning-by-exporting proposes that firms improve after entering international markets because exposure to foreign customers, competitors and technologies generates knowledge and organisational learning. Freixanet and Federo (2023), reviewing 167 studies across four decades, identify institutional conditions, firm resources, capabilities and internationalisation strategy as important elements, while noting fragmentation over the precise mechanisms behind post-entry gains. Evidence from UK micro, small and medium firms suggests self-selection and learning-by-exporting can operate simultaneously, with innovation mediating the relationship (Gkypali et al., 2021). This matters for interpretation here: a predictive association between innovation and exporter status does not demonstrate that innovation causes exporting, since innovation may raise the probability of export entry, exporting may stimulate innovation, or both may occur simultaneously. The machine learning and SHAP results in this thesis are accordingly interpreted as evidence of predictive association rather than causal effect. Together, RBV, heterogeneous-firm theory and self-selection/learning-by-exporting justify expecting observable differences between exporters and non-exporters, supporting the use of firm-level characteristics to predict export participation.

2.3 Firm-Level Determinants of SME Export Participation

A substantial empirical literature identifies firm characteristics associated with export participation. Although the magnitude and significance of individual determinants vary across

countries and sectors, the literature consistently highlights firm size, productivity, access to finance, innovation and technology, quality certification, ownership and human capital.

2.3.1 Firm Size and Productivity

Firm size is among the most consistently identified correlates of exporting: larger firms generally have greater production capacity, financial resources and organisational capability to absorb the fixed costs of foreign-market entry, and Bernard et al. (2007) show exporters tend to be larger and more productive, consistent with heterogeneous-firm theory. Productivity relates closely to this advantage, since more productive firms can absorb export-related costs while remaining price-competitive internationally. However, productivity and size are not interchangeable: smaller firms with specialised products, strong innovation or digital market access may internationalise despite limited scale, so their effects need consideration jointly with other characteristics.

2.3.2 Access to Finance

Access to finance is another major determinant, since exporting requires working capital for production, certification, transportation, marketing and the delay between shipment and payment; credit-constrained firms may be unable to exploit otherwise viable opportunities. Chowdhury et al. (2022) find a positive relationship between external finance and labour productivity among Bangladeshi SMEs, while Phan et al. (2022) find credit constraints negatively affect export decisions, moderated by institutional conditions. Finance is particularly important in Africa: Beck and Cull (2014) document substantial financing constraints, and Wellalage and Locke (2016) link informality to access to external credit in Sub-Saharan Africa, suggesting financial variables may carry substantial predictive importance in the EAC. Digital financial infrastructure may also matter. Konte and Tetteh (2023) find mobile money complements rather than replaces traditional financial services for African firms, a dimension that may distinguish the region from European exporter-prediction settings.

2.3.3 Innovation, Technology and Quality Certification

Innovation and technological capability are frequently linked to export participation through improved productivity, product quality and ability to meet international requirements, though the strength of this relationship varies by context. Teplova et al. (2022) find innovative activity and R&D facilitate market entry in emerging markets; Mmadubuko et al. (2025), using WBES data on Indian SMEs, similarly find product innovation positively related to export performance,

strengthened by quality certification. Certification may reduce information asymmetry between domestic producers and international buyers by signalling reliability Pacheco et al. (2022) find positive relationships between ISO certification and internationalisation among Portuguese industrial SMEs, particularly low- and medium-technology firms. Mwatu (2022), using Kenyan WBES data, finds certification, trade finance and employee training associated with export performance. However, the relationship is context-dependent Panda and Sharma's (2023) meta-regression finds domestic innovation has a stronger positive export effect in developed than developing economies, cautioning against assuming determinants from high-income economies carry equivalent weight in the EAC.

2.3.4 Ownership, Human Capital and Managerial Capability

Ownership characteristics can affect internationalisation through access to capital, technology, knowledge and international networks; foreign-owned firms may benefit from existing distribution networks that lower barriers to export entry. Gender ownership may also be relevant, though its relationship with exporting can reflect differences in resource access rather than gender itself — Ackah et al. (2024) find an observed productivity gap between male- and female-owned Ghanaian businesses disappears once financial constraints are accounted for. Human capital and managerial capability matter similarly, since exporting requires knowledge of foreign customers, regulations, logistics and competitive conditions that employee training and managerial experience can support, although managerial capability is difficult to measure directly in secondary surveys, so available variables often function as imperfect proxies. Overall, export participation results from interactions among multiple resources and capabilities rather than any single characteristic, justifying predictive approaches capable of evaluating variables jointly and capturing nonlinear interactions.

2.4 Machine Learning Approaches to Export Prediction

Predicting whether a firm exports is a supervised binary-classification problem. Logistic regression is an important baseline for its transparency and established use, but flexible machine learning methods can identify nonlinearities and interactions that traditional regression would need to specify explicitly. Random Forest (Breiman, 2001) combines multiple decision trees built on bootstrap samples and randomly selected predictor subsets, aggregating predictions to reduce the instability of individual trees while retaining their capacity for nonlinear pattern detection.

Bayesian Additive Regression Trees (BART) represent an outcome as the combined contribution of multiple regression trees within a Bayesian framework; BART-MIA extends this by letting missingness enter the splitting mechanism directly (Sparapani et al., 2021), potentially valuable for WBES data where item non-response varies across variables and countries.

A further issue is class imbalance: exporters are frequently a minority of surveyed firms, so a classifier can achieve superficially high accuracy by disproportionately predicting non-exporters. Chawla et al. (2002) introduced the Synthetic Minority Over-sampling Technique (SMOTE), generating synthetic minority observations from neighbouring minority cases; oversampling must be restricted to training data to avoid leakage into evaluation. Recent gradient-boosting algorithms such as XGBoost (Chen and Guestrin, 2016), LightGBM and CatBoost can perform strongly on structured data, but model selection here follows the benchmarking objective: changing the model set against an established exporter-prediction study would weaken comparability, since performance differences could reflect algorithm choice rather than economic context. This study accordingly prioritises comparability with the anchor study (Section 2.5).

2.4.1 Explainable Artificial Intelligence and SHAP

Predictive accuracy alone is insufficient when outputs are meant to inform policy, since complex models may classify accurately without transparently explaining why a given firm receives a high or low export probability. Explainable Artificial Intelligence (XAI) addresses this through methods for interpreting model behaviour; the most influential, SHapley Additive exPlanations (SHAP), introduced by Lundberg and Lee (2017), uses Shapley values from cooperative game theory to decompose a prediction into contributions from individual predictors. SHAP supports both global explanations (which characteristics matter most across the sample) and local explanations (why an individual firm receives a given prediction) relevant to export-promotion applications requiring both overall understanding and firm-specific justification for targeted intervention. SHAP nevertheless explains the fitted model rather than the underlying causal process, and its outputs can be affected by correlated predictors; SHAP values are therefore interpreted here as measures of predictive contribution rather than causal evidence.

2.5 Anchor Study: Micocci and Rungi (2023)

The principal methodological benchmark for this thesis is Micocci and Rungi's (2023) *Predicting Exporters with Machine Learning*, published in *World Trade Review*, which demonstrates that

exporter status can be predicted from firm-level information and provides a direct precedent for treating export participation as a supervised classification problem. Using ORBIS financial-statement data for French firms observed 2010–2018, the authors compare five approaches: logistic regression, LASSO-logit, classification trees, Random Forest and BART-MIA, with BART-MIA producing the strongest results (accuracy up to 0.90) and constructing continuous exporting scores shown to be robust under alternative definitions of exporter status. The study's importance lies not simply in applying machine learning but in evaluating multiple classifiers on a common dataset and evaluation framework, attributing performance differences to modelling approach rather than inconsistent samples; its practical orientation—connecting exporting scores to firms' financial resources for trade-promotion and trade-credit targeting—shows how firm-level predictions can move beyond academic classification toward decision support.

The study nevertheless sets a demanding benchmark for application to the EAC. ORBIS financial statements from France differ substantially from WBES survey observations in variable availability, measurement and missingness, and France is a high-income economy with mature financial and institutional systems, unlike Kenya, Uganda and Tanzania. This thesis therefore does not reproduce the French dataset; instead, it transfers the underlying exporter-prediction framework to a different economic and data environment while maintaining comparability where possible, testing whether Micocci and Rungi's central proposition that exporter status is sufficiently structured to be predicted from firm characteristics, holds within an East African SME context.

2.6 SME Exporting in Sub-Saharan Africa and the East African Community

The Sub-Saharan African context introduces factors that may affect both exporter prevalence and the predictive importance of firm characteristics. First, informality remains substantial: Wellalage and Locke (2016) show informality and financial constraints are closely related in Sub-Saharan Africa, and because WBES data principally represent formal enterprises, findings should not be generalised automatically to informal microenterprises. Second, access to finance is a persistent constraint: Beck and Cull (2014) document substantial financing barriers, and while informal finance can partly compensate for limited formal lending, its capacity to support larger investments is limited (Mpofu and Sibindi, 2022). Third, regional integration has not removed all cross-border barriers: the EAC Customs Union reduced internal barriers and established a common external

tariff, yet infrastructure, logistical and non-tariff barriers continue to constrain regional trade (Hersi et al., 2025). At the continental level, the African Continental Free Trade Area (AfCFTA) offers further market-access opportunities, and Geda and Yimer (2023) estimate substantial potential increases in intra-African exports from tariff reductions, though this depends on domestic investment, institutional quality and firm capability. The WTO (2016) identifies trade finance, information and trade-related skills as significant SME barriers, and UNCTAD (2024) highlights continuing constraints on intra-African trade-reinforcing that EAC export behaviour sits in a substantially different environment from the anchor study's European context. This contextual difference is central to the present research, which empirically evaluates the predictive relevance of firm characteristics using harmonised regional data rather than assuming high-income-economy models transfer unchanged.

2.7 Empirical Evidence and Closely Related Work

The broader empirical literature on SME exporting remains dominated by explanatory studies. African research in particular has generally investigated individual determinants of export performance through econometric or structural-equation approaches rather than developing out-of-sample exporter classifiers. For Kenya, Mwatu (2022) uses WBES data and econometric techniques to identify certification, trade finance and training as relevant characteristics; in Tanzania, Ringo et al. (2022) examine entrepreneurial orientation among manufacturing-exporting SMEs, finding relationships between innovativeness, risk-taking and export performance. Such studies provide valuable contextual evidence but are designed for hypothesis testing rather than prediction. Haddoud et al. (2018) offer a methodologically relevant African example, examining Algerian SME export propensity through an extended RBV framework and fuzzy c-means clustering, showing firms can be separated into resource profiles with different export propensities though clustering differs fundamentally from supervised classification, since it is not trained to maximise predictive performance against observed status. The gap is visible beyond export research too: Kazimoto and Baadel (2025) apply supervised machine learning to fintech data from informal Lesotho merchants to predict loan default, demonstrating the feasibility of firm-level African classification tools despite a different predicted outcome.

African SME research contains substantial evidence on determinants of firm behaviour and an emerging machine-learning literature for business prediction, but the two strands have rarely been

integrated to predict observed SME exporter status using harmonised multi-country African firm data. The present study occupies this intersection, using observed export participation rather than a constructed readiness measure, supervised classification rather than determinant-by-determinant hypothesis testing, harmonised WBES data across selected EAC economies, and explainable machine learning to interpret predictive contributions.

2.8 Synthesis of the Empirical Literature

The literature demonstrates that SME export participation is shaped by a combination of firm-level resources, capabilities and contextual conditions. A relatively consistent set of characteristics emerges as relevant across studies firm size, productivity, access to finance, innovation and technology, quality certification, ownership and human capital though their relative importance is not necessarily constant across countries, industries or institutions. RBV attributes systematic differences between exporters and non-exporters to variation in firm resources (Barney, 1991), heterogeneous-firm theory to differences in productivity and capacity to overcome fixed export costs (Melitz, 2003), and empirical evidence confirms exporters tend to be larger, more productive and more resource-intensive (Bernard et al., 2007). Export participation, however, cannot be explained by size or productivity alone; financial capacity, innovation, certification, ownership and human capital jointly shape a multidimensional firm profile.

Relationships established in developed economies cannot be assumed to hold with equivalent strength in developing countries, given differences in financial-system development, institutional quality, infrastructure, informality and technological capability, particularly relevant in Sub-Saharan Africa (Beck and Cull, 2014; Wellalage and Locke, 2016). Kenya, Uganda and Tanzania share a regional integration framework but continue to differ in financial development, infrastructure and firm characteristics, so predictors cannot be assumed to carry identical importance across the three economies. Methodologically, much SME internationalisation research has focused on explanation rather than prediction; Micocci and Rungi (2023) mark an important departure by showing firm-level characteristics can predict exporter status through machine learning, but their benchmark rests on French ORBIS data, leaving open whether comparable relationships hold using survey-based SME information from substantially different environments. Table 2.1 compares the studies most closely related to the present research.

Table 2.1: Comparison of Closely Related Studies and the Present Research

Study	Context and Data	Method	Key Contribution/Relevance	Limitation/Gap Relative to the Present Study
Micocci and Rungi (2023)	French firms; ORBIS financial-statement data, 2010–2018	Logit, LASSO-logit, CART, Random Forest and BART-MIA	Principal anchor study; demonstrates that exporter status can be predicted using firm-level information and provides an international performance benchmark	Conducted in a single high-income European economy using financial-statement data; transferability to survey-based EAC SMEs remains untested
Mwatu (2022)	Kenya; WBES data	Tobit and random-effects GLS	Provides directly relevant Kenyan evidence concerning firm-level characteristics associated with export performance	Single-country explanatory analysis; does not construct or evaluate an out-of-sample exporter classifier
Ringo et al. (2022)	Tanzanian manufacturing SMEs	PLS-SEM	Provides EAC evidence concerning firm capabilities and export performance	Single-country primary survey and hypothesis-testing design rather than supervised exporter prediction
Hansen-Addy et al. (2023)	27 African countries; WBES firm-level observations	Panel regression and propensity-score methods	Demonstrates the value of large-scale WBES data for investigating African firm performance and institutional conditions	Focuses on regulatory effects and general firm performance rather than exporter-status prediction
Konte and Tetteh (2023)	African firms; financial and mobile-money characteristics	Panel regression	Highlights the relevance of conventional and digital financial access to African firm performance	Explanatory productivity analysis rather than exporter classification; does not benchmark predictive performance
Kazimoto and Baadel (2025)	Informal merchants in Lesotho; fintech transaction data	Logistic regression and Support Vector Machines	Demonstrates the feasibility of supervised firm-level classification in an African business context	Predicts loan default rather than export status and uses proprietary single-context data
Present study	SMEs in Kenya, Uganda and Tanzania; harmonised WBES data	Supervised classification, out-of-sample evaluation and SHAP-based interpretation	Transfers exporter prediction to a multi-country EAC context and evaluates the predictive contribution of firm characteristics	Addresses the methodological, contextual, comparative and interpretability gaps identified in the literature

(Source: developed by the researcher from the studies reviewed in this chapter.)

This comparison highlights a divide between African evidence, which is largely explanatory, and the exporter-prediction literature, which is predictive but developed in a different institutional and data environment. The present research investigates whether the predictive logic established internationally can transfer to harmonised East African firm-survey data, and underscores that policy application requires understanding which characteristics drive predictions, not merely a classifier that distinguishes exporters from non-exporters.

2.9 Research Gap and Contribution of the Study

The broad determinants of SME export participation are already well established, so the core research problem is not to identify another list of associated variables. Open questions instead remain about whether these characteristics can be combined into reliable out-of-sample predictions of exporter status within the EAC, whether predictive performance established in a developed-country setting transfers to this context, and whether the resulting predictions can be interpreted usefully for export-promotion decision-making. Four interconnected gaps follow.

The methodological gap concerns the field's explanatory orientation: conventional econometrics suits hypothesis testing but is not designed to maximise predictive performance on unseen data. A variable can be statistically significant without contributing much to out-of-sample classification, while variables with complex nonlinear effects may contribute meaningfully to prediction even when hard to specify in linear models. Machine learning evaluates characteristics jointly and captures such interactions without pre-specification; Micocci and Rungi (2023) demonstrate this feasibility, but the limited number of exporter-prediction studies means further evidence is needed on its robustness across different firm populations. This study addresses the gap by treating export participation explicitly as a supervised classification problem evaluated out-of-sample.

The contextual and geographical gap reflects that influential firm-level export research is concentrated in developed economies, the anchor study relies on French firms via ORBIS. Institutional and economic conditions in Kenya, Uganda and Tanzania differ considerably in financing, infrastructure, transaction costs, digital financial adoption and trade barriers, so predictive relationships from a developed-country environment should not be assumed to generalise. This study tests exporter prediction specifically within the three EAC economies.

The data and comparative gap concerns comparability: African SME studies often rely on individual-country surveys with different questionnaire designs and sampling, complicating cross-country comparison. The WBES's standardised instruments allow common variables to be harmonised across Kenya, Uganda and Tanzania, enabling both pooled analysis and examination of country-level differences, since regional membership does not imply firm-level homogeneity.

The interpretability and practical gap reflects that improved accuracy alone does not make a model useful for policy: agencies need to understand why a firm receives a given predicted probability, and a black-box prediction offers limited guidance. SHAP addresses this by decomposing fitted predictions into feature-level contributions at global and local levels (Lundberg and Lee, 2017); this study treats these explanations as descriptions of how the fitted model uses available characteristics, not as causal estimates.

Table 2.2 aligns the thesis's four research questions with these gaps.

Table 2.2: Alignment of Research Questions with Identified Literature Gaps

Research Question	Principal Literature Gap Addressed	Purpose within the Study
RQ1 Predictive features	Limited evidence regarding the relative predictive importance of established firm-level export determinants within the EAC	Identifies which firm-level resources and characteristics contribute most strongly to exporter-status predictions
RQ2 Benchmark comparison	Limited evidence testing established exporter-prediction approaches outside high-income European financial-statement settings	Evaluates predictive performance in the EAC context relative to the international anchor study
RQ3 Practical predictive application	African SME research predominantly explains export-related outcomes rather than constructing validated exporter-classification tools	Assesses whether firm-level information can support sufficiently reliable exporter predictions for potential decision-support applications
RQ4 Regional consistency	Limited harmonised evidence regarding whether exporter predictors remain consistent across EAC economies	Examines similarities and differences in predictive patterns across Kenya, Uganda and Tanzania

(Source: developed by the researcher from the literature review and research questions.)

Collectively, these gaps define the dissertation's contribution: not to establish that firm characteristics matter for exporting, which extensive evidence already supports, but to investigate whether observable firm-level resources can be combined into reliable, interpretable predictions of SME exporter status within the EAC, how that performance compares with an established international benchmark, and whether the underlying characteristics remain consistent across Kenya, Uganda and Tanzania.

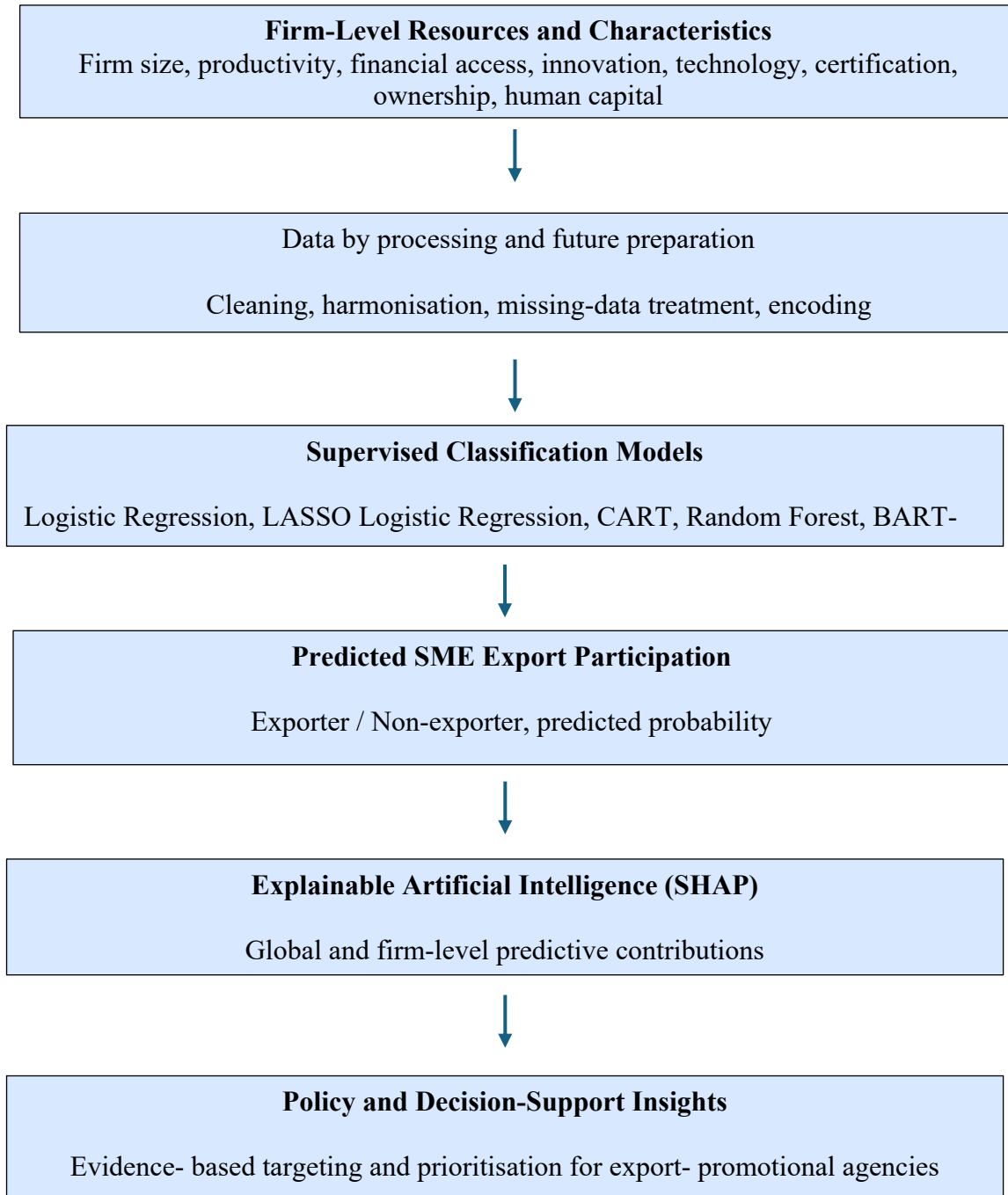
2.10 Conceptual Framework

The conceptual framework connects the proposition that exporters possess systematically different resources and capabilities with the empirical task of predicting exporter status from observable characteristics. Rather than specifying causal pathways from individual variables to export participation, it treats firm characteristics as information that may help distinguish exporters from non-exporters, with machine-learning algorithms determining how that information combines to generate out-of-sample predictions. It draws on RBV and heterogeneous-firm theory for its firm-level foundation, on the determinants literature for its measurable predictor domain (firm size, productivity, financial access, innovation, technology, certification, ownership and human capital), on machine learning as the combining mechanism, and on explainable AI as its interpretive component.

The framework comprises five interconnected stages: (1) firm-level resources and characteristics the observable predictors determined by theoretical relevance, prior evidence and availability across Kenya, Uganda and Tanzania; (2) data pre-processing and feature preparation cleaning, harmonising, treating missing data and encoding variables per the strategy in Chapter 3; (3) supervised machine-learning modelling, in which competing classifiers estimate exporter status and are compared for the strongest balance of discrimination, robustness and interpretability, not accuracy alone; (4) predicted SME export participation, the classification or probability output reflecting how closely a firm's characteristics resemble training-data exporters; and (5) explainable artificial intelligence and interpretation, applying SHAP to identify which characteristics contribute most to predictions and how these vary across observations and countries. The final implication concerns policy and decision support: if exporter status can be predicted reliably out-of-sample and interpreted transparently, such models may offer complementary evidence for

export-promotion institutions, conditional on observed performance and without implying causal pathways or automatic policy eligibility.

Figure 2.1 Conceptual framework



(Source: Developed by the researcher based on Barney(1991),Melitz(2003),Breiman(2001),Lundberg and Lee(2017), and Micocci and Rungi(2023).

CHAPTER 3. RESEARCH METHODOLOGY

3.1 Introduction

This chapter presents the research methodology used to investigate and predict SME export participation within the East African Community (EAC). Building on Chapter 2, the methodology determines whether observable firm-level resources and characteristics can distinguish exporting from non-exporting SMEs, focusing on Kenya, Uganda and Tanzania using secondary firm-level data from the World Bank Enterprise Surveys (WBES).

The study adopts a quantitative, cross-sectional, predictive design based on supervised machine learning. Export participation is treated as a binary classification outcome, with competing models evaluated on their ability to predict exporter status for observations not used in training. Following Micocci and Rungi (2023), the principal international benchmark, the analysis compares five classification approaches: Logistic Regression (LOGIT), LASSO-regularised Logistic Regression (LOGIT-LASSO), Classification and Regression Trees (CART), Random Forest (RF), and Bayesian Additive Regression Trees with Missingness Incorporated in Attributes (BART-MIA).

Particular attention is given to challenges of cross-country firm-level survey data variable harmonisation, missing observations and class imbalance. Because exporters are a minority, performance is assessed using multiple classification metrics rather than accuracy alone, and the original class distribution is retained in the test data. Explainable Artificial Intelligence (XAI), specifically SHapley Additive exPlanations (SHAP), is subsequently used to interpret the contribution of firm-level characteristics to model predictions.

3.2 Research Philosophy, Approach and Design

3.2.1 Research Philosophy and Approach

The study adopts a positivist philosophy and a quantitative approach, appropriate because it relies on observable, measurable firm-level data and systematic computational procedures rather than subjective interpretation. This is consistent with the WBES, which provides numerical and categorical information on employment, ownership, finance, innovation, technology, certification and trade activity that can be processed using statistical and machine-learning techniques.

The study follows a predominantly deductive orientation: predictor selection and interpretation are informed by established theory, including the Resource-Based View (Barney, 1991) and Melitz's (2003) heterogeneous-firm framework, both of which provide a basis for expecting observable differences between exporters and non-exporters. However, the primary objective is prediction rather than causal inference the study does not claim that a characteristic causes exporting, only that combinations of observable characteristics can predict exporter status. Relationships identified by the models and subsequent SHAP analysis are therefore interpreted as predictive associations rather than causal effects, since characteristics such as finance, innovation, certification and technology may both influence and be influenced by export participation.

3.2.2 Empirical Research Design

The study is empirical research, developing and evaluating machine-learning models on observed real-world firm-level data rather than theoretical propositions or simulation. This involves preparing and harmonising WBES data from Kenya, Uganda and Tanzania, constructing the exporter-status outcome, training alternative algorithms and evaluating predictive performance on held-out observations.

The design is cross-sectional and uses secondary data: the WBES provides structured firm-level information directly relevant to the framework and research questions established in Chapters 1 and 2, and no new information is collected from SME owners or managers. Because firms are observed at a single survey round rather than tracked over time, the study identifies predictive patterns associated with exporter status at the observed period rather than estimating longitudinal changes. The analysis sits within a supervised machine-learning classification framework, with export participation as a binary outcome and observable firm characteristics as predictors, trained on a designated sample and evaluated on held-out test observations to assess generalisation.

3.2.3 Benchmarking Research Component

The study also incorporates a benchmarking component, central to the dissertation because the predictive framework is evaluated against Micocci and Rungi (2023). Rather than choosing algorithms by contemporary popularity, the study adopts their five model classes, providing a systematic basis for comparing predictive behaviour between the EAC application and the international benchmark. The exercise considers both predictive performance and relative model ranking, examining whether approaches that perform strongly in the benchmark, particularly

BART-MIA retain their advantage on WBES data from Kenya, Uganda and Tanzania. It is not treated as an exact replication, since the two settings differ in geographical context, firm population, data source and available predictors; the purpose is to evaluate transferability of the methodological framework, and any performance differences are interpreted with these contextual differences in mind.

3.2.4 Overall Research Design

Overall, the study is a quantitative empirical design incorporating cross-sectional secondary-data analysis, supervised machine learning and methodological benchmarking. The empirical component provides evidence from observed EAC firm-level data, while benchmarking evaluates the resulting models against an established exporter-prediction framework. SHAP complements this design by interpreting the model's predictions, allowing the research to examine not only whether SME exporter status can be predicted accurately but also which firm characteristics contribute most strongly to those predictions.

3.3 Data Source, Study Population and Sample

3.3.1 Data Source

The study uses secondary firm-level data from the WBES, which provides structured information on private-sector establishments across multiple economies, covering firm characteristics, ownership, employment, finance, innovation, technology, quality certification, business environment and international trade activity, closely matching the determinants of SME export participation identified in Chapter 2. The WBES's standardised framework facilitates comparative analysis across the selected EAC economies, reducing comparability problems that would arise from combining independently designed national surveys.

The analysis uses WBES datasets for Kenya, Uganda and Tanzania, processed initially at country level before comparable variables are identified, harmonised and pooled; only variables that can be operationalised with sufficient conceptual and measurement consistency across surveys are retained, since differing variable names or coding structures do not necessarily imply differing underlying concepts. Using observed rather than simulated data provides an appropriate basis for assessing whether exporter status can be predicted using characteristics realistically observable by policymakers, financial institutions or export-promotion agencies.

3.3.2 Study Population and SME Definition

The target population consists of formally registered SMEs in the selected WBES datasets, with the establishment as the unit of observation. SME status is operationalised by employment size: small enterprises (5–19 permanent full-time employees) and medium-sized enterprises (20–99 employees). Establishments with 100 or more employees are excluded. A common employment-based definition is important in a multi-country study because national legal SME definitions may differ by employment, turnover, assets or sector; applying one rule across the three countries prevents observed differences from merely reflecting different national classifications. The resulting population should not be interpreted as representing every small business in the three economies, it reflects the formal establishments captured by the WBES sampling frame, and informal microenterprises fall outside the scope of the predictive framework.

3.3.3 Country Selection

The study covers Kenya, Uganda and Tanzania, a relevant East African setting for testing whether a common machine-learning framework can predict SME export participation across different national business environments. Their inclusion allows the analysis to move beyond a single-country exercise: a pooled dataset can identify regional patterns while retaining a country identifier that permits examination of national differences, since institutional conditions, financial access and technology adoption may operate differently across these economies. The three countries are therefore used in two complementary ways combined for a pooled EAC prediction framework, and compared to assess whether predictor importance and model performance vary by country, directly supporting the comparative research question. The term "EAC" is used carefully throughout: the dataset does not cover every current member state, so findings should be read as evidence from the selected countries rather than the whole Community.

3.3.4 Sample Construction

Sample construction follows a sequential procedure applying the same eligibility criteria across datasets. Country datasets are first imported and inspected separately to identify variables needed for SME classification, exporter-status construction and predictor development, examining labels, response categories and missing-value codes before pooling. The employment criterion (5–99 employees) is then applied to identify eligible SMEs, and the dependent variable is constructed from WBES information on firms' international sales, classifying eligible SMEs consistently as

exporters or non-exporters per Section 3.4. Predictor variables are then assessed for cross-country comparability, with variables that cannot be harmonised reliably excluded rather than retained under inconsistent definitions; where equivalent concepts are represented differently, common indicators are constructed from the underlying response categories. Finally, the cleaned country-level datasets are combined with a retained country identifier, enabling pooled estimation alongside comparison of predictive patterns between countries.

Two analytical samples are constructed: a full-sector SME sample as the principal dataset, and a manufacturing-only SME sample used for robustness and for direct benchmarking against the exporter-prediction literature. Final sample sizes and exporter/non-exporter distributions are reported in Chapter 4 based on the actual processed data, once SME filtering, exporter-status construction, harmonisation and data-quality procedures are complete.

3.4 Variable Definition and Operationalisation

Variables are selected on the basis of the theoretical framework and empirical evidence reviewed in Chapter 2, together with their availability and comparability across the selected WBES datasets. The analysis distinguishes the target variable, SME export participation, from a set of firm-level predictors representing resources, capabilities and structural characteristics. Because the study is predictive rather than explanatory, variables are treated as observable features from which the models learn patterns, not as causal determinants; the final predictor set includes only variables reliably harmonised across Kenya, Uganda and Tanzania.

3.4.1 Target Variable: Export Participation

The target variable is SME export participation, a binary outcome. WBES records the percentage of an establishment's sales across domestic markets, direct exports and indirect exports through third parties, providing the basis for classification: export participation equals 1 if direct exports are greater than zero or indirect exports are greater than zero, and 0 otherwise. The study focuses on export participation rather than export intensity. predicting whether an SME participates in international markets, not the value or share of export sales. A single harmonised definition is applied across the three countries so exporter status carries the same interpretation throughout the pooled dataset; the exact WBES variable(s) and classification threshold are confirmed during data preparation once the coding structures of the three datasets have been inspected, to avoid specifying a definition unsupported by the actual data.

3.4.2 Predictor Variables

Predictor selection is theory-informed rather than purely data-driven, drawing on the characteristics identified in Chapter 2 as the initial candidate set. A predictor is retained only where a conceptually comparable measure can be constructed across all three WBES datasets, since inconsistent measurement could produce artificial country differences. Candidate groups include: firm size (permanent full-time employment) and age (years since commencement); access to finance, harmonised across countries' question wording; innovation capability (product/process innovation and R&D, where available); technology and digital capability (foreign-licensed technology, website use); quality certification; ownership characteristics, particularly foreign ownership; human and managerial capability (training or managerial experience, where consistently available); and sector and country indicators, capturing differences in export opportunity across industries and between Kenya, Uganda and Tanzania.

3.4.3 Variable Harmonisation

Harmonisation is central to operationalisation: the three datasets are not assumed identical merely because they share a survey programme. Variable labels, question wording, coding categories, units and missing-value structures are compared before pooling, following three principles — variables must represent the same underlying concept across countries; response categories must be directly comparable or recodable into a common definition; and variables with substantial structural incompatibility are excluded rather than forced into the pooled dataset. All harmonisation decisions are documented in the analytical code to ensure reproducibility.

3.4.4 Operationalisation Framework

Table 3.1: Operationalisation Framework

Construct	Role	Operational Measure
Export participation	Target	Binary exporter/non-exporter classification from WBES sales-destination variables
Firm size	Predictor	Permanent full-time employees
Firm age	Predictor	Years since operations began
Access to finance	Predictor	Harmonised WBES finance-access/constraint measure
Innovation	Predictor	Comparable product/process innovation measure
R&D	Predictor	Comparable R&D activity/expenditure measure, if available
Technology	Predictor	Comparable technology-adoption measure
Digital capability	Predictor	Website/digital-use indicator, if consistently available
Quality certification	Predictor	Internationally recognised certification
Foreign ownership	Predictor	Comparable foreign-ownership measure
Human/managerial capability	Predictor	Training/managerial measure, where comparable
Sector	Predictor/control	Harmonised sector classification
Country	Predictor/control	Kenya, Uganda or Tanzania

(Source: developed by the researcher based on the theoretical framework and WBES variable structure.)

3.5 Data Preparation and Exploratory Data Analysis

Preparation of the WBES data precedes model development to ensure the three country datasets are sufficiently comparable and suitable for supervised learning, involving inspection, harmonisation, sample filtering, construction of the dependent variable, treatment of missing observations, encoding of categorical variables and transformation of numerical variables where required. Exploratory Data Analysis (EDA) then examines the resulting sample and identifies data-quality issues before model training.

3.5.1 Data Inspection and Harmonisation

The three country datasets are examined separately before merging, establishing the number of observations, available variables, coding schemes and missing-value patterns, with particular attention to variables needed for SME classification, exporter-status construction and predictor operationalisation. Because identical variable names are not assumed to represent identical measures across country survey rounds, labels, wording, response categories and units are compared, and equivalent concepts represented by different codes are recoded into a common definition. Variables that cannot be reliably harmonised are excluded, prioritising measurement consistency over the number of available predictors; a country identifier is retained after merging.

3.5.2 Sample Filtering and Data Cleaning

Following harmonisation, datasets are filtered to the 5-99 employee SME definition, then examined for duplicates, invalid responses, inconsistent coding and implausible values, with survey-specific missing-value codes distinguished from valid responses and special categories such as "don't know" assessed according to question meaning rather than treated as ordinary numerical values. The exporter-status variable is then constructed and checked for consistent classification across the three countries.

3.5.3 Missing Data Treatment

Missing observations are examined systematically, since item non-response varies across firms, variables and countries; the extent and pattern of missingness determine whether a variable requires imputation, exclusion or model-specific treatment. Where imputation is needed, it is performed within the modelling pipeline using training data only, so no information from held-out test observations is used. BART-MIA receives particular consideration because its missingness-incorporated tree structure allows missing predictor information to enter the model directly rather than requiring removal or conventional imputation — one methodological reason for retaining it in the model comparison. Variables with excessive or structurally inconsistent missingness that cannot be made sufficiently comparable across countries are removed rather than manufactured through extensive imputation.

3.5.4 Encoding and Transformation

Categorical variables are converted into numerical representations: binary variables are coded consistently across countries, and nominal variables such as sector and country use dummy or one-hot encoding where required. Continuous variables are examined for skewness, extreme observations and scale differences, with highly skewed variables transformed where appropriate; transformations are documented in the final analytical code. Scaling is applied selectively — LASSO logistic regression requires attention to predictor scale since its penalty can otherwise be influenced by measurement units, while tree-based methods (CART, Random Forest, BART-MIA) do not require standardisation. All preprocessing that estimates parameters from the data is performed after the train-test split, within the training pipeline.

3.5.5 Prevention of Data Leakage

Preventing data leakage, where information from observations meant to remain unseen influences training, producing artificially optimistic performance estimates is essential to the design. The test sample is created before any operation that could use information from the full dataset, including imputation, scaling, feature selection and SMOTE, all of which are fitted only within training data; test observations remain untouched until models are fully developed. This ensures the evaluation in Section 3.7 represents genuine out-of-sample performance.

3.5.6 Exploratory Data Analysis

EDA examines the distribution of firms across Kenya, Uganda and Tanzania, sectoral composition, firm-size categories and the prevalence of exporters and non-exporters, using descriptive statistics for numerical variables and frequency distributions for categorical variables, and comparing exporters and non-exporters descriptively. The dependent variable's distribution receives particular attention given expected imbalance: preliminary data indicate exporters represent approximately 13.3% of the pooled full-sector sample and 22.4% of the manufacturing-only sample, meaning a classifier could achieve apparently high accuracy simply by predicting the majority non-exporter class. This informs the class-imbalance strategy in Section 3.6 and the use of multiple performance metrics in Section 3.7. EDA also examines missingness, outliers and relationships between predictors, purely to inform preprocessing and modelling decisions rather than to establish causal relationships.

3.6 Model Development and Class-Imbalance Strategy

Following data preparation, five supervised models are developed to predict SME export participation, LOGIT, LOGIT-LASSO, CART, RF and BART-MIA selected to maintain comparability with Micocci and Rungi (2023).

3.6.1 Training and Test Data

The final dataset is divided into training and test sets using stratified sampling, ensuring exporters and non-exporters remain represented in approximately the same proportions in both subsets. Training data are used for preprocessing, training, tuning and class-imbalance treatment; test data are reserved solely for final out-of-sample evaluation.

3.6.2 Classification Models

LOGIT is the primary statistical baseline for its simplicity and interpretability, estimating the probability an SME belongs to the exporter class from the selected characteristics. LOGIT-LASSO extends this with an L1 regularisation penalty, shrinking less informative predictors toward zero and handling potentially correlated or redundant variables. CART provides a nonlinear alternative, recursively dividing observations by decision rules and capturing interactions without pre-specification. RF combines predictions from multiple decision trees, with randomisation of observations and predictors reducing the instability and overfitting of individual trees while capturing complex nonlinear relationships. BART-MIA, the most flexible model, combines multiple Bayesian regression trees while its Missingness Incorporated in Attributes component allows missing predictor information to enter the tree structure directly particularly relevant to WBES data and consistent with the anchor benchmark.

3.6.3 Class Imbalance

Given the expected imbalance (approximately 13.3% exporters in the full-sector sample, 22.4% in the manufacturing sample), class weighting and/or SMOTE are considered during training where appropriate, generating synthetic minority-class observations rather than simply duplicating existing ones. Any resampling is performed only on training data; the test set is never oversampled and retains the original class distribution, ensuring final performance measures reflect the actual observed data.

3.7 Model Evaluation and Benchmarking

Model performance is evaluated on the held-out test dataset to determine how well each classifier predicts exporter status for unseen observations. Because exporters are a minority class, accuracy alone could mislead.

3.7.1 Evaluation Metrics

The study evaluates models using accuracy, balanced accuracy (average of sensitivity and specificity), precision, recall/sensitivity, specificity, F1-score, ROC-AUC and PR-AUC (particularly useful for the imbalanced exporter class), alongside confusion matrices showing true/false positives and negatives. No single metric determines the best model in isolation; particular attention is given to recall, F1-score, PR-AUC and balanced accuracy, since correctly identifying the minority exporter class is central to the research objective.

3.7.2 Model Comparison

The five models are evaluated on the same test observations using comparable procedures, with performance compared across both the full-sector and manufacturing-only samples. The best-performing model is determined from overall performance across the evaluation metrics rather than accuracy alone, allowing the study to identify whether more flexible models such as RF and BART-MIA meaningfully improve on the LOGIT, LOGIT-LASSO and CART benchmarks.

3.7.3 Benchmarking Against Micocci and Rungi (2023)

Results are compared with Micocci and Rungi's (2023) findings, focusing on whether the relative ranking of the five algorithms remains similar when transferred from French firm-level data to EAC firm-survey data; the manufacturing-only analysis provides an additional basis for comparison. This is treated as a methodological benchmark rather than exact replication, since datasets, countries, firm populations and available variables differ, addressing whether the exporter-prediction framework demonstrates similar predictive usefulness in the EAC context rather than assuming the French results automatically transfer.

3.8 Model Interpretability and SHAP Analysis

Predictive accuracy alone does not explain why a model classifies an SME as an exporter, so Explainable Artificial Intelligence via SHAP is incorporated. SHAP measures the contribution of individual predictors to model predictions, providing both global explanations (ranking variables by overall contribution across the dataset, using mean absolute SHAP values) and local explanations (how characteristics contribute to a particular firm's prediction). SHAP plots can also show the direction of contributions for example, whether particular predictor values increase or decrease predicted export probability. SHAP is applied to the best-performing model from Section 3.7, and the resulting feature-importance rankings are compared against the theoretical expectations from Chapter 2 and, where appropriate, across the three countries. SHAP values are interpreted as model-based predictive contributions rather than causal effects: high SHAP importance indicates a variable plays an important role in how the fitted model distinguishes exporters from non-exporters, not that changing it would cause a firm to export. This combines predictive performance with interpretability, supporting the practical objective of a transparent SME export-prediction framework.

3.9 Reliability, Ethical Considerations and Methodological Limitations

3.9.1 Reliability and Reproducibility

Reliability is supported through a consistent, reproducible analytical procedure across the three countries, with documented computational steps for data preparation, harmonisation, training and evaluation. A fixed random seed improves reproducibility, training and test datasets are kept separate, and preprocessing and resampling are applied only to training data to prevent leakage. All five models are evaluated using the same test data and metrics, providing a consistent basis for comparison and benchmarking against Micocci and Rungi (2023).

3.9.2 Ethical Considerations

The research uses secondary WBES data with no direct interaction with human participants and no primary data collection. Analysis is conducted for academic purposes without identifying individual firms or respondents. Model predictions are interpreted responsibly as statistical estimates rather than definitive assessments of future export success, and SHAP results are treated as indicators of predictive importance rather than causality; the resulting model is intended to support, not replace, professional judgement in SME export-promotion decisions.

3.9.3 Methodological Limitations

Several limitations should be recognised. The study relies on secondary WBES data, so the analysis is restricted to variables the surveys collect, and factors such as managerial capability, international networks and strategic orientation may not be fully captured. The cross-sectional design limits causal interpretation: models identify associations, not causation. The three surveys are not necessarily conducted in identical years, so differing economic conditions across periods may influence results, and the countries differ in institutional, financial, technological and sectoral conditions that country indicators cannot fully capture. Class imbalance remains an important limitation, since SMOTE and class-weighting cannot fully compensate for limited minority-class information. Finally, findings are specific to the selected EAC countries and survey periods; strong performance here does not guarantee equivalent performance elsewhere, and external validation using additional countries or future WBES rounds would be needed before operational deployment.

CHAPTER 4. RESULTS AND ANALYSIS

4.1 Introduction

This chapter presents the empirical results on SME export participation in Kenya, Tanzania and Uganda, following the design in Chapter 3. It evaluates the ability of five supervised models — Logistic Regression (LOGIT), LASSO Logistic Regression (LOGIT-LASSO), Classification and Regression Trees (CART), Random Forest (RF), and Bayesian Additive Regression Trees with Missingness Incorporated in Attributes (BART-MIA) to distinguish exporting from non-exporting SMEs, selected to maintain comparability with Micocci and Rungi (2023).

Results are presented in stages: the final analytical sample and the distribution of exporter status and missing predictor information across the three countries; results from model development and cross-validation; out-of-sample evaluation using a comprehensive set of classification metrics (accuracy, balanced accuracy, precision, recall, specificity, F1-score, ROC-AUC and PR-AUC a broad set adopted because exporters are a minority class, so accuracy alone would be an incomplete guide); and finally country-level differences in performance, including BART-MIA relative to the alternative models and the international benchmark. The chapter addresses the four research questions through complementary evidence on predictive features, benchmark performance, practical prediction and regional consistency.

4.2 Descriptive Results and Sample Characteristics

After data inspection, harmonisation and quality procedures, the final full-sector analytical sample comprises 2,229 SME observations: 1,024 for Kenya, 600 for Tanzania and 605 for Uganda, each with twelve harmonised predictors and the binary exporter-status outcome.

Table 4.1: Final Sample and Exporter Distribution by Country

Country	Observations	Exporters	Non-exporters	Exporter (%)
Kenya	1,024	179	845	17.48
Tanzania	600	86	504	14.58
Uganda	605	66	539	10.91
Total	2,229	331	1,898	14.85

(Source: author's analysis of World Bank Enterprise Survey data.)

Kenya had the highest observed exporter participation, followed by Tanzania and Uganda, but exporters were a minority in every country most markedly in Uganda, where they made up roughly one-tenth of the sample. This confirms an imbalanced classification task, meaning a model could achieve high accuracy simply by favouring the majority non-exporter class; balanced accuracy, precision, recall, F1-score, ROC-AUC and PR-AUC are therefore reported alongside accuracy throughout. Kenya also provides the largest analytical sample (over 1,000 observations, more than twice as many exporters as Uganda), while Tanzania and Uganda each contain around 600 relevant when interpreting country-level recall and precision, which are calculated from relatively few positive observations per test sample. Overall, export participation is uncommon in all three countries, and the degree of imbalance differs between them, providing the basis for the model-performance assessment that follows.

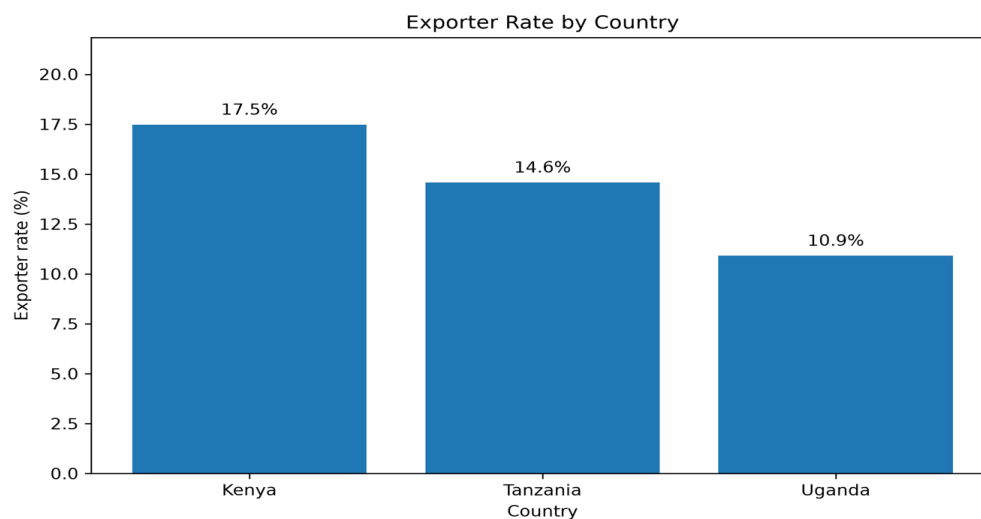


Figure 4.1: Exporter Rate by Country

4.3 Data Quality and Missingness

Missingness across the outcome and twelve predictors varied considerably by country: Kenya showed the lowest overall levels, while Tanzania and Uganda showed higher levels for several predictors. The largest concentration was in the d2 variable — 87 missing observations (14.50%) in Tanzania and 46 (7.60%) in Uganda, versus substantially fewer in Kenya — and in b7, with 29 (4.83%) missing in Tanzania and 22 (3.64%) in Uganda compared with 3 (0.29%) in Kenya. Other predictors showed considerably lower missingness (e.g., Tanzania's e6 at 1.50%, b3a and c39 at 0.67%; Uganda's k30 and b3a below 1%; Kenya limited to single-digit missing counts for b7, h1 and e6). The exporter outcome itself was complete for Kenya and Uganda; Tanzania had 10

missing observations (1.67%), distinguished from missing-predictor cases during quality assessment.

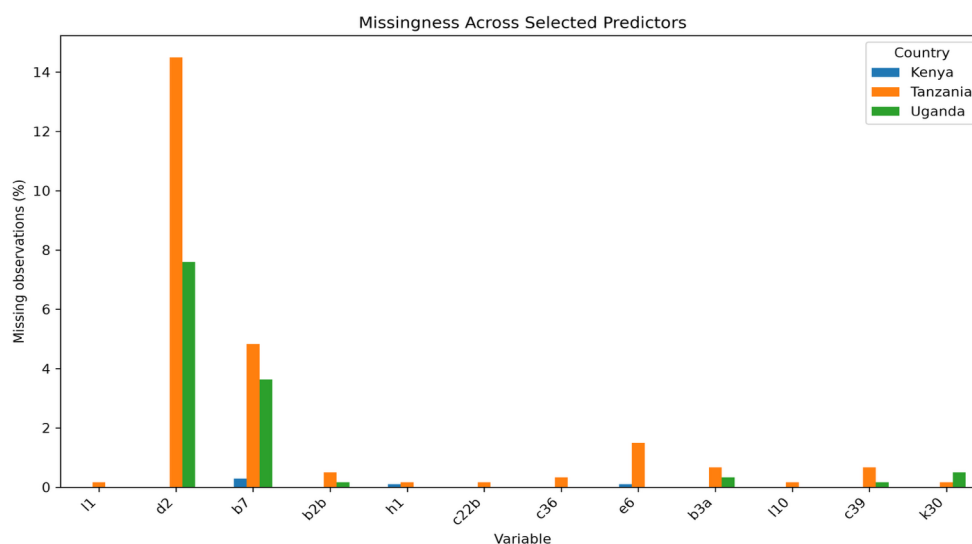


Figure 4.2: Missingness Across Selected Predictors

Table 4.2: Summary of Missingness in the Final Country Samples

Country	Final N	Missing exporter
Kenya	1,024	0
Tanzania	600	10
Uganda	605	0

Tanzania showed the greatest overall presence of missing predictor information (143 observations affected), followed by Uganda (75) and Kenya (5). Missing values were not treated as evidence of invalid observations; missingness was addressed through the modelling strategy set out in Chapter 3. The data-quality assessment confirmed that the majority of observations remained available in each country (raw and clean sample sizes unchanged) and that missingness was concentrated in a limited number of predictors rather than spread uniformly patterns subsequently incorporated into model preparation and validation. Final checks confirmed the twelve predictors and binary outcome were structurally consistent across all three countries.

4.4 Data Integrity and Modelling Sample Validation

Validation procedures confirmed the datasets retained their expected structure. Raw-to-clean reconciliation of the c39 variable matched expected counts in all three countries (Kenya: 482 coded

0, 542 coded 1, no missing; Tanzania: 484 coded 0, 112 coded 1, 4 missing; Uganda: 480 coded 0, 124 coded 1, 1 missing), and structural validation confirmed all twelve predictors and the target variable were present, with clean sample sizes unchanged from the raw data. A duplicate-profile check found one potential duplicate in Kenya and two in Tanzania (none in Uganda); these were retained, since identical values across modelling variables alone do not establish that observations are duplicate firms, and removing them risked losing valid data.

Table 4.3: Duplicate Modelling-Profile Assessment

Country	Potential duplicate rows	Decision
Kenya	1	Retain
Tanzania	2	Retain
Uganda	0	Retain

4.4.1 Train–Test Modelling Sample

The validated datasets were split into training and test partitions (approximately 80/20, stratified): test sets of 205 (Kenya), 118 (Tanzania) and 121 (Uganda) observations, with corresponding training samples of 819, 472 and 484. Test data played no role in model selection, providing an independent basis for final evaluation.

4.5 Model Preparation and Training

The five models were estimated separately for each country, so that country-specific differences in performance could be retained rather than assuming uniform performance across the region. LOGIT and LOGIT-LASSO provided linear benchmarks (the latter with regularisation), CART and Random Forest provided tree-based non-linear alternatives, and BART-MIA provided a Bayesian non-linear model capable of incorporating missingness directly. Training and cross-validation used only the training observations; test observations remained untouched until final evaluation, allowing assessment of whether strong development-stage performance transferred to unseen firms.

The four traditional models were fitted successfully for all three countries (12 fitted models in total) and loaded successfully for evaluation. Cross-validation using ROC-AUC as the principal measure was conducted during development, ahead of the broader test-set evaluation reported below. BART-MIA was likewise trained separately per country, and validation confirmed its test predictions matched each test set's size exactly (Table 4.4).

Table 4.4: BART-MIA Test Prediction Validation

Country	Test observations	Test predictors	Predicted exporters	Observed exporters
Kenya	205	12	14	36
Tanzania	118	12	3	17
Uganda	121	12	6	13

Predicted exporter counts were substantially lower than observed counts in every country, foreshadowing the high specificity and low recall BART-MIA displays in the full test-set evaluation below. At this stage no model is designated as superior; performance is determined from the untouched test observations, since strong cross-validation performance does not necessarily carry over to unseen firms.

4.6 Overall Out-of-Sample Model Performance

Table 4.5: Out-of-Sample Predictive Performance by Country and Model (Accuracy / Balanced Accuracy / Precision / Recall / Specificity / F1 / ROC-AUC / PR-AUC)

Country	Model	Accuracy	Bal. Acc.	Precision	Recall	Specificity	F1	ROC-AUC	PR-AUC
Kenya	Logistic Regression	0.7659	0.705	0.3929	0.6111	0.7988	0.4783	0.7678	0.5505
Kenya	LASSO Logistic Regression	0.7659	0.705	0.3929	0.6111	0.7988	0.4783	0.7676	0.5503
Kenya	CART	0.7951	0.7009	0.4348	0.5556	0.8462	0.4878	0.7927	0.4371
Kenya	Random Forest	0.8293	0.6997	0.5143	0.5	0.8994	0.507	0.8489	0.5579
Kenya	BART-MIA	0.8634	0.6439	0.7857	0.3056	0.9822	0.44	0.8526	0.6266
Tanzania	Logistic Regression	0.7119	0.636	0.2571	0.5294	0.7426	0.3462	0.7123	0.2914
Tanzania	LASSO Logistic Regression	0.6695	0.6357	0.2381	0.5882	0.6832	0.339	0.6925	0.2741
Tanzania	CART	0.7881	0.6316	0.3182	0.4118	0.8515	0.359	0.6453	0.2781
Tanzania	Random Forest	0.8559	0.6223	0.5	0.2941	0.9505	0.3704	0.7472	0.4265
Tanzania	BART-MIA	0.8475	0.5195	0.3333	0.0588	0.9802	0.1	0.7746	0.4324
Uganda	Logistic Regression	0.7686	0.6674	0.2414	0.5385	0.7963	0.3333	0.7593	0.4261
Uganda	LASSO Logistic Regression	0.7686	0.6674	0.2414	0.5385	0.7963	0.3333	0.7593	0.4261
Uganda	CART	0.8099	0.5552	0.1875	0.2308	0.8796	0.2069	0.6268	0.1625
Uganda	Random Forest	0.8512	0.646	0.3333	0.3846	0.9074	0.3571	0.7906	0.3437
Uganda	BART-MIA	0.8926	0.6015	0.5	0.2308	0.9722	0.3158	0.7735	0.4086

No single model achieves the highest value across all measures; the results instead reveal different trade-offs between identifying exporters and correctly classifying non-exporters.

In Kenya, BART-MIA achieved the highest accuracy (0.8634), precision (0.7857), specificity (0.9822), ROC-AUC (0.8526) and PR-AUC (0.6266), but its recall of 30.56% was well below Random Forest's 50.00% and the logistic models' 61.11% high ROC-AUC alongside high specificity and low recall indicates strong ranking ability but a conservative classification threshold. Random Forest's lower accuracy came with a higher F1-score (0.5070 vs 0.4400) and higher recall, while the logistic models had the highest recall but lower precision and accuracy; Kenya provides BART-MIA's strongest results overall.

In Tanzania, Random Forest had the highest accuracy (0.8559), marginally ahead of BART-MIA (0.8475), and the highest precision among traditional models (0.5000). BART-MIA had the highest ROC-AUC (0.7746) and PR-AUC (0.4324) but a recall of just 0.0588 (specificity 0.9802), while Random Forest's recall of 0.2941 and F1-score of 0.3704 far exceeded BART-MIA's F1 of 0.1000; the logistic models achieved higher balanced accuracy and recall than Random Forest at the cost of specificity and accuracy. This shows that strong probability-based discrimination need not translate into strong binary classification of the minority class at the chosen threshold.

In Uganda, BART-MIA had the highest accuracy (0.8926) and specificity (0.9722), but Random Forest had the highest balanced accuracy (0.6460) and ROC-AUC (0.7906). The logistic models had the highest recall (0.5385) but low precision (0.2414); Random Forest gave an intermediate profile (recall 0.3846, precision 0.3333, specificity 0.9074) and the highest F1-score (0.3571) among the five models, while BART-MIA's PR-AUC (0.4086) remained relatively strong. The model with the highest overall accuracy was again not the one with the strongest balance between classes.

Across countries, BART-MIA achieved the highest ROC-AUC and PR-AUC in Kenya and Tanzania, while Random Forest led in Uganda on both measures among the top models (logistic models led PR-AUC in Uganda). Random Forest was comparatively the most consistent performer (ROC-AUC of 0.8489, 0.7472 and 0.7906 in Kenya, Tanzania and Uganda respectively, versus BART-MIA's 0.8526, 0.7746 and 0.7735). No algorithm is universally superior: the preferred model depends on which criterion e.g., minimising false non-exporter identification versus

maximising exporter identification is prioritised, since BART-MIA in particular tends toward high specificity and strong ROC-AUC alongside comparatively low recall.

4.7 ROC-AUC Results

ROC-AUC assesses discrimination between exporters and non-exporters across classification thresholds, independent of any single threshold. In Kenya, BART-MIA led narrowly over Random Forest (0.8526 vs 0.8489), with CART at 0.7927 and the logistic models around 0.767 both leading models show strong discrimination. In Tanzania, BART-MIA again led (0.7746), followed by Random Forest (0.7472), Logistic Regression (0.7123), LASSO Logistic Regression (0.6925) and CART (0.6453); notably, despite the highest ROC-AUC, BART-MIA's recall at the selected threshold was very low, showing its ranking ability exceeded its binary classification performance. In Uganda, Random Forest led (0.7906), followed by BART-MIA (0.7735), the logistic models (0.7593) and CART (0.6268) the only country where Random Forest, not BART-MIA, gave the strongest discrimination.

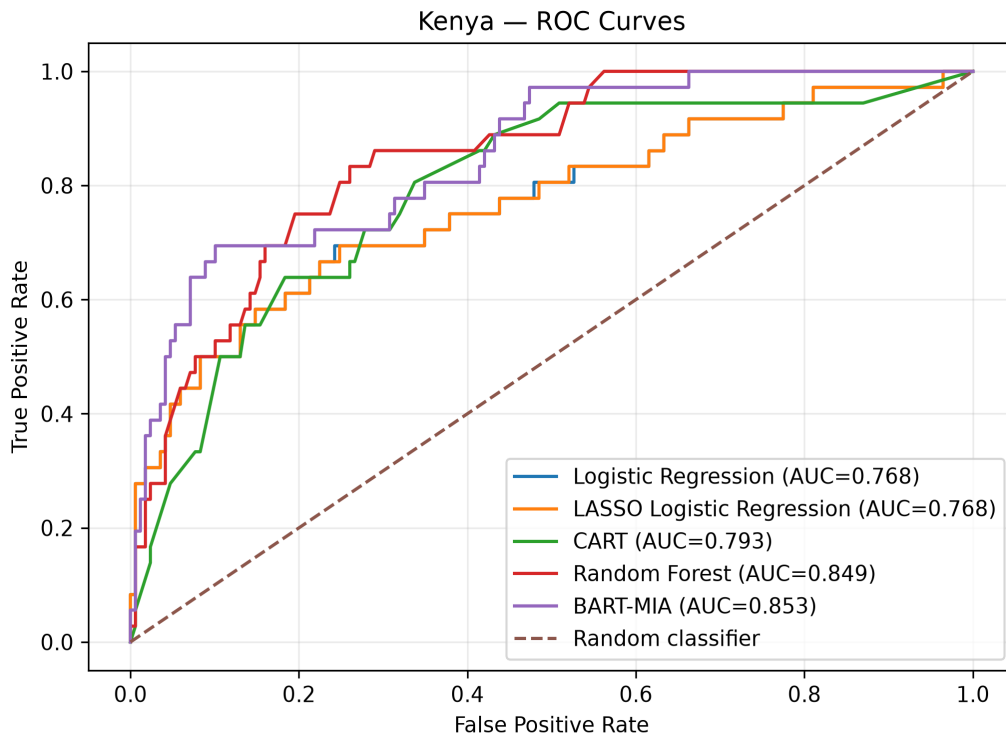


Figure 4.3: ROC Curves for Kenya

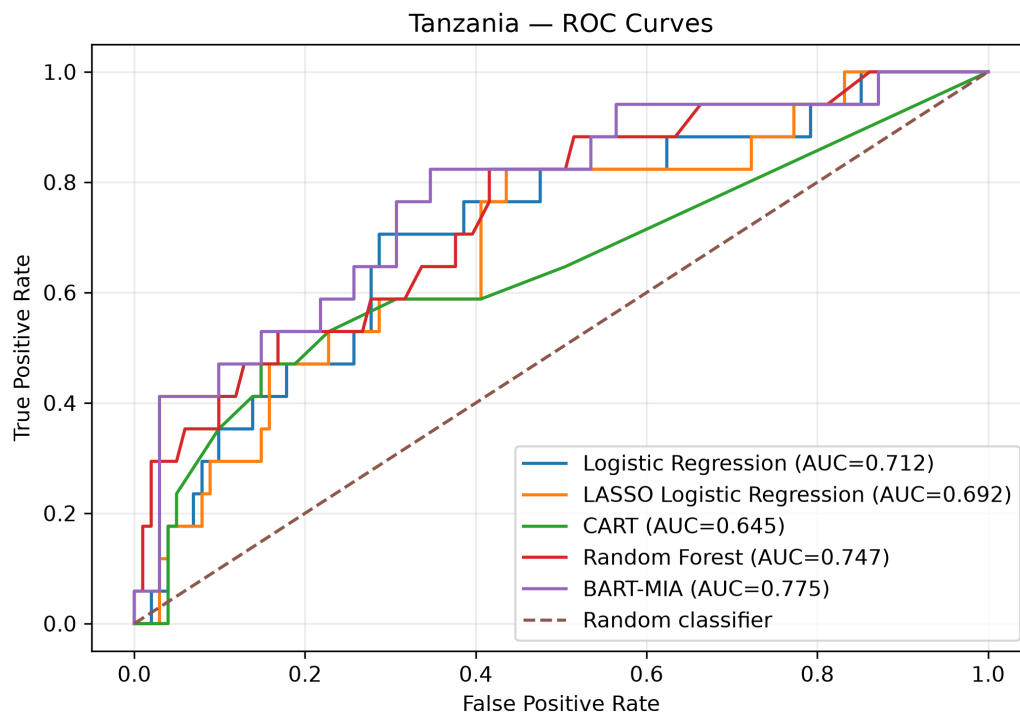


Figure 4.4: ROC Curves for Tanzania

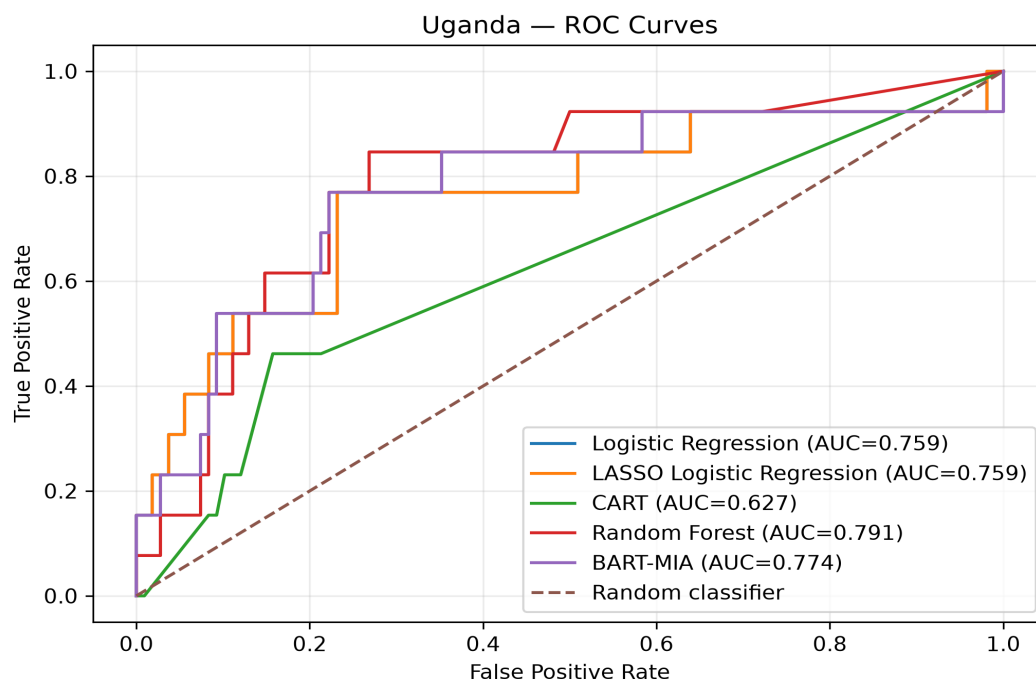
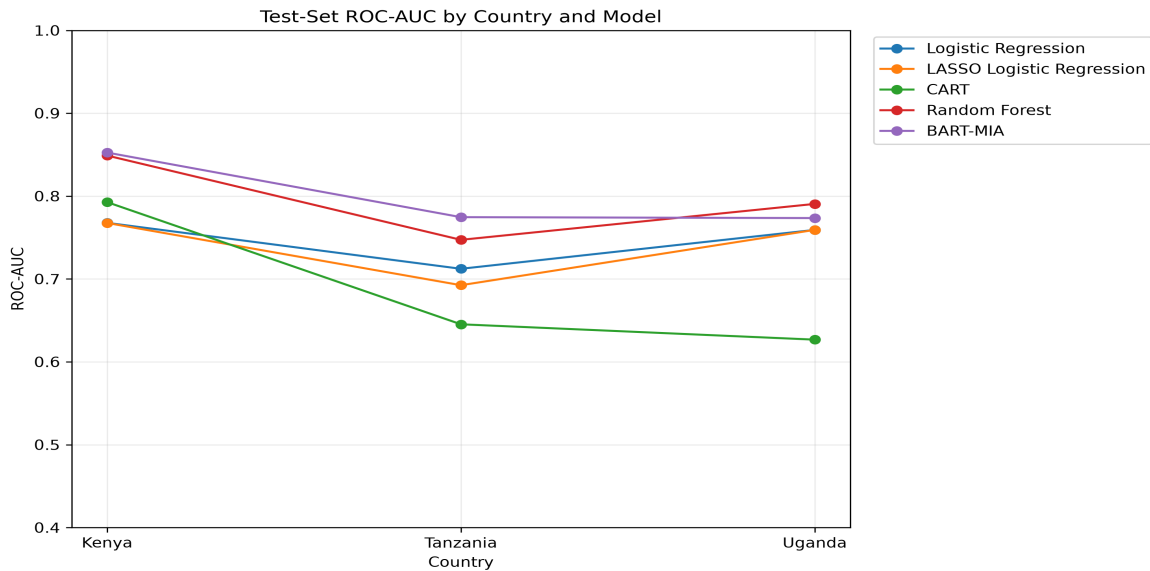


Figure 4.5: ROC Curves for Uganda

Table 4.6: Highest Test ROC-AUC by Country

Country	Highest-performing model	Test ROC-AUC
Kenya	BART-MIA	0.8526
Tanzania	BART-MIA	0.7746
Uganda	Random Forest	0.7906

**Figure 4.6: Test-Set ROC-AUC by Country and Model**

The predictive relationship between firm characteristics and exporter status is therefore not identical across the three countries, and the same model's discrimination varies by country (Random Forest: 0.7472–0.8489; BART-MIA: 0.7735–0.8526) evidence that performance is shaped by the country-specific modelling environment. Because ROC-AUC evaluates ranking rather than performance on the small exporter class specifically, precision–recall analysis is examined next.

4.8 Precision–Recall Results

PR-AUC focuses more directly on the precision–recall relationship for the exporter class. In Kenya, BART-MIA led (PR-AUC 0.6266), followed by Random Forest (0.5579) and the logistic

models (~ 0.550); CART was lowest (0.4371). BART-MIA's higher PR-AUC is consistent with its high precision (0.7857) but low recall (0.3056), while Random Forest's more balanced precision (0.5143) and recall (0.5000) gave it a higher F1-score despite a lower PR-AUC. In Tanzania, BART-MIA again led (0.4324), closely followed by Random Forest (0.4265), with the logistic models and CART between 0.27 and 0.29; BART-MIA's precision (0.3333) and very low recall (0.0588) versus Random Forest's precision (0.5000) and recall (0.2941) show that BART-MIA's higher PR-AUC reflects its overall threshold-independent behaviour rather than stronger exporter identification at the chosen threshold. In Uganda, the two logistic models led (PR-AUC 0.4261, with the highest recall at 0.5385 but precision of only 0.2414), followed by BART-MIA (0.4086, precision 0.5000, recall 0.2308), Random Forest (0.3437) and CART (0.1625).

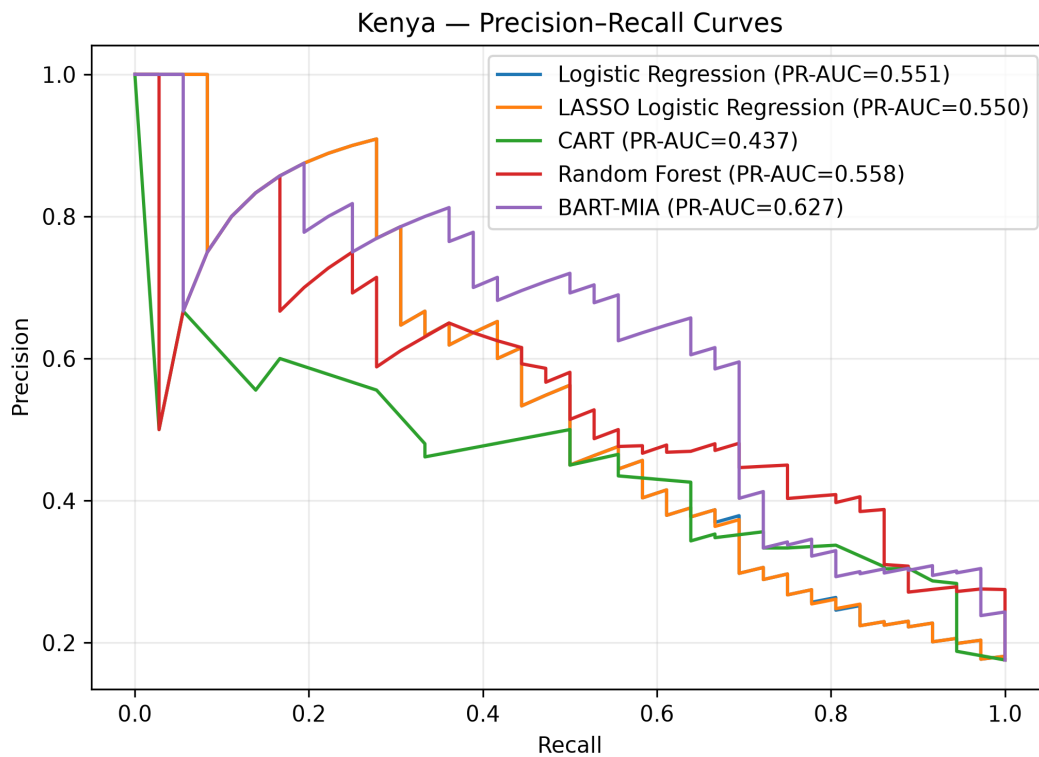


Figure 4.7: Precision–Recall Curves for Kenya

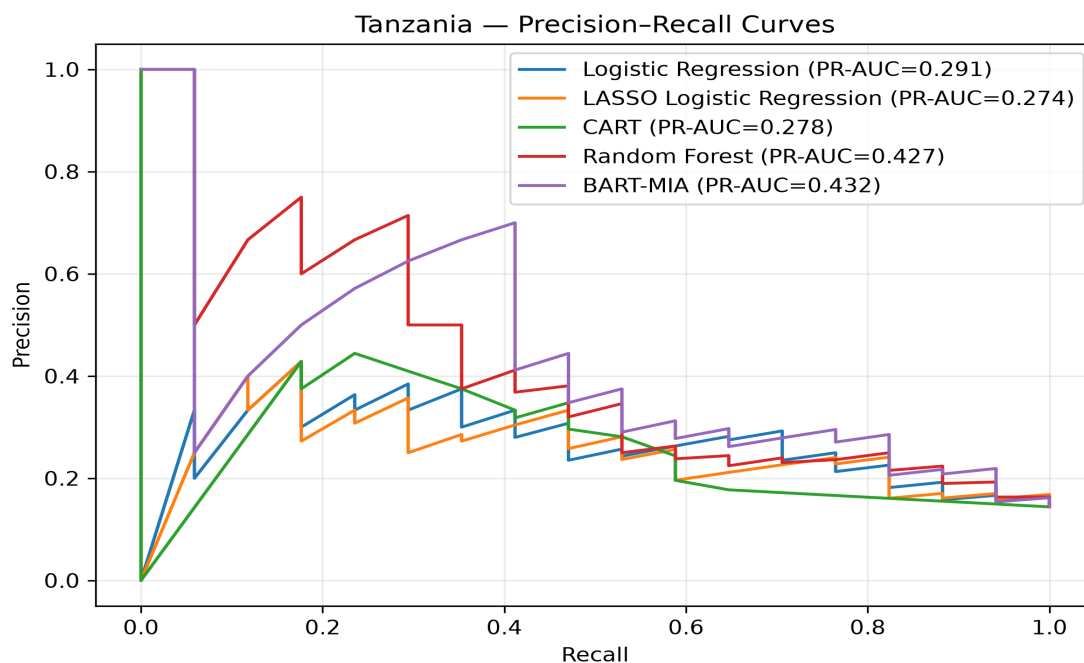


Figure 4.8: Precision-Recall Curves for Tanzania

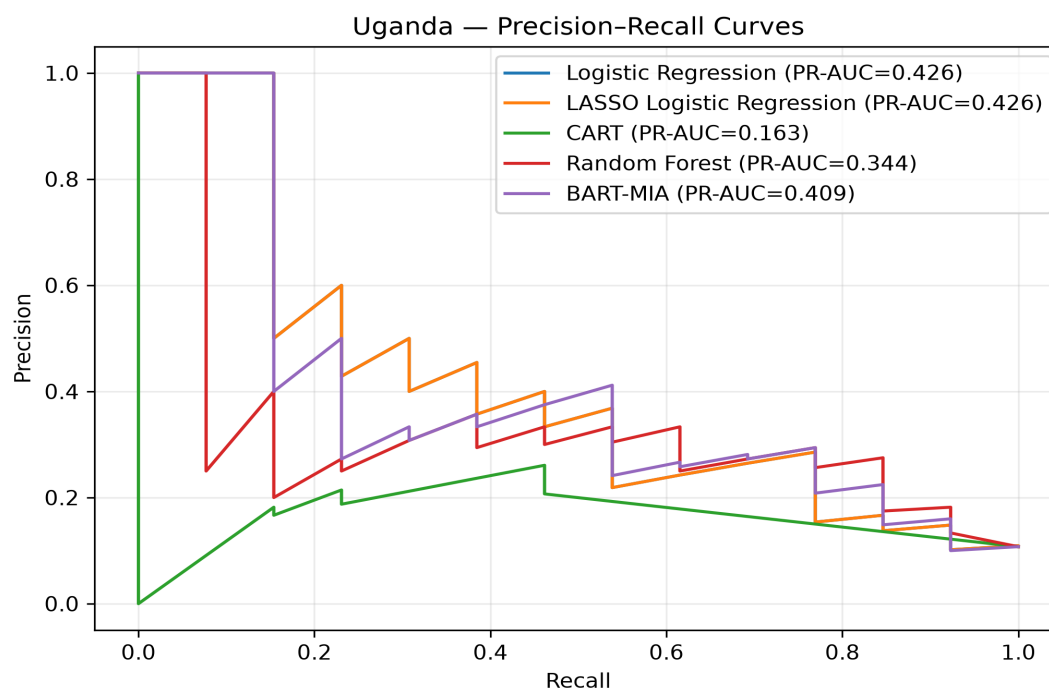


Figure 4.9: Precision-Recall Curves for Uganda

Table 4.7: Highest Test PR-AUC by Country

Country	Highest-performing model	Test PR-AUC
Kenya	BART-MIA	0.6266
Tanzania	BART-MIA	0.4324
Uganda	Logistic Regression / LASSO Logistic Regression	0.4261

These results confirm why model selection cannot rest on a single metric: in Kenya BART-MIA led PR-AUC and ROC-AUC but Random Forest gave a higher F1-score and recall; in Tanzania BART-MIA led both AUC measures but had extremely low recall; in Uganda the logistic models led PR-AUC and recall while Random Forest led ROC-AUC. The models differ not only in overall accuracy but in their ability to sustain precision while identifying the minority exporter class.

4.9 Confusion-Matrix Results

Confusion matrices for the preferred conventional model, Random Forest chosen for its strongest overall conventional-model performance across the three countries provide a direct view of classification trade-offs. In Kenya, Random Forest correctly classified 152 non-exporters and 18 exporters, misclassifying 17 non-exporters as exporters and 18 exporters as non-exporters (recall 0.5000, specificity 0.8994, F1 0.5070, accuracy 0.8293) a comparatively balanced profile next to BART-MIA's high specificity but much lower recall. In Tanzania, it correctly classified 96 non-exporters and 5 exporters (5 false positives, 12 false negatives; recall 0.2941, specificity 0.9505, accuracy 0.8559, the highest of the five models) a reminder that high accuracy here reflects the large number of correctly classified non-exporters more than success identifying exporters. In Uganda, it correctly classified 98 non-exporters and 5 exporters (10 false positives, 8 false negatives; recall 0.3846, specificity 0.9074, accuracy 0.8512, balanced accuracy 0.6460, F1 0.3571) again a relatively balanced profile compared with BART-MIA's more conservative classification.

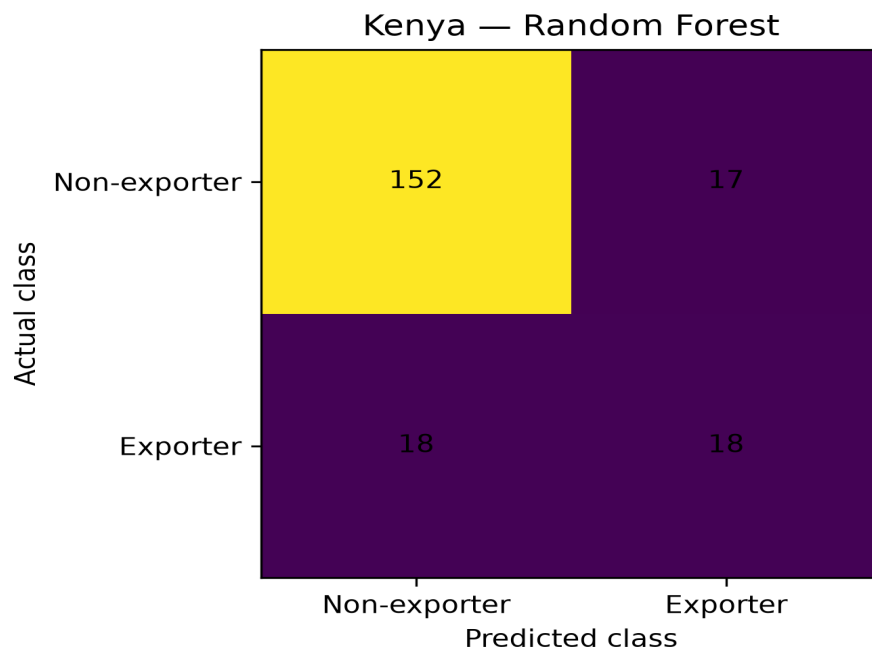


Figure 4.10: Kenya Confusion Matrix: Random Forest

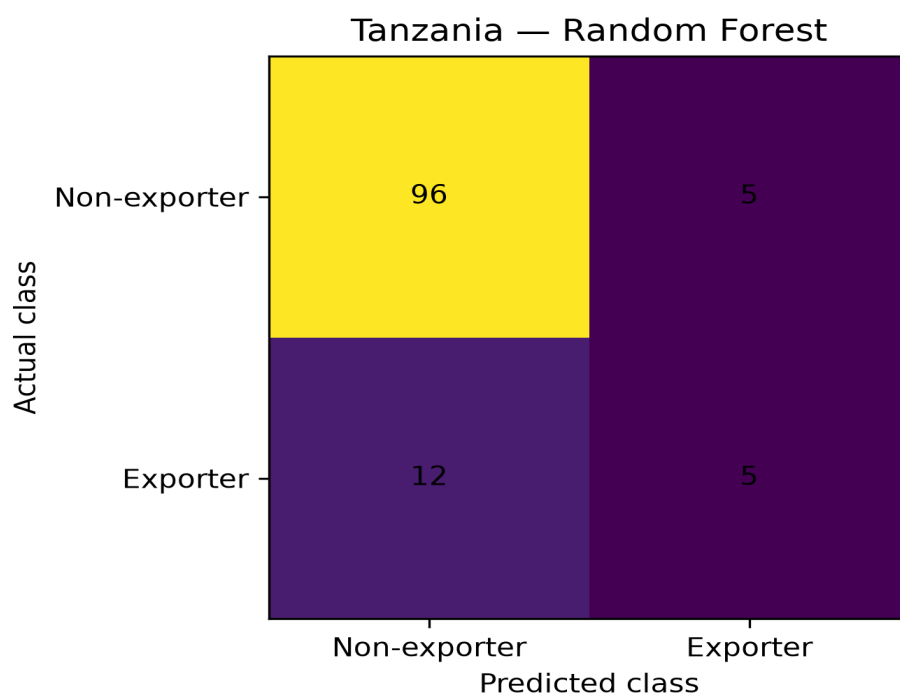


Figure 4.11: Tanzania Confusion Matrix: Random Forest

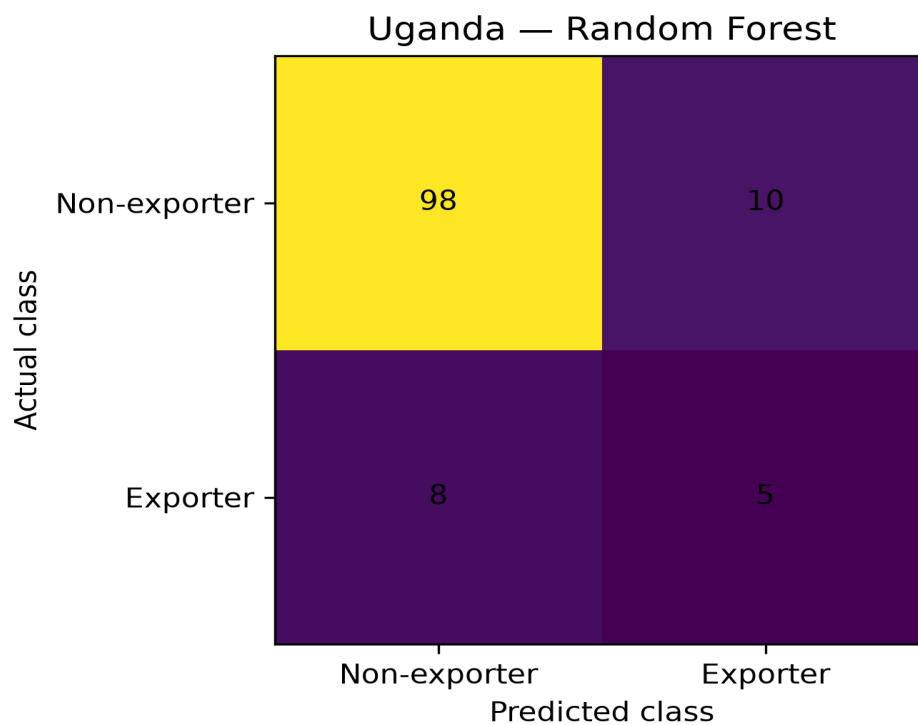


Figure 4.12: Uganda Confusion Matrix: Random Forest

Country	Correct non-exporters	Incorrect non-exporters	Correct exporters	Incorrect exporters
Kenya	152	17	18	18
Tanzania	96	5	5	12
Uganda	98	10	5	8

Random Forest identified exporters most successfully in Kenya (18 correctly classified, recall 0.5000), with five correctly classified in each of Tanzania (recall 0.2941) and Uganda (recall 0.3846). Correctly classified non-exporters were substantially more numerous in every country, reflecting both strong specificity and the much larger number of non-exporting firms — underscoring that high overall accuracy can still coincide with a substantial share of exporters going unidentified.

4.10 Cross-Validation and Test-Set Performance

Comparing cross-validation with test-set performance indicates how well the models generalise. For Random Forest, which achieved the strongest cross-validated ROC-AUC in all three countries, cross-validation scores were consistently higher than test-set scores.

Table 4.8: Random Forest Cross-Validation and Test-Set ROC-AUC

Country	Model	CV ROC-AUC	Test ROC-AUC	Test – CV
Kenya	Random Forest	0.9574	0.8489	−0.1085
Tanzania	Random Forest	0.9733	0.7472	−0.2261
Uganda	Random Forest	0.9905	0.7906	−0.1999

The largest gap was in Tanzania, followed by Uganda and Kenya; test-set ROC-AUC nonetheless remained above 0.74 in all three, indicating the models retained real discriminatory capability out-of-sample, albeit lower than cross-validation suggested. This generalisation gap, most pronounced in Tanzania and Uganda, reinforces the decision to treat the untouched test sets, rather than cross-validation scores as the primary basis for model comparison, since they were not used in fitting or tuning.

4.10.1 Implications for Model Performance Assessment

Model performance depends substantially on the evaluation criterion used. Random Forest led on cross-validated ROC-AUC in all three countries but showed lower test-set performance; BART-MIA led test ROC-AUC in Kenya and Tanzania, Random Forest in Uganda. High accuracy did not consistently mean high recall BART-MIA's accuracy advantage in Kenya and Uganda came with much lower recall than several alternatives, and it led PR-AUC in Kenya and Tanzania but not Uganda. Exporter prediction is feasible, but predictive performance is sensitive to both country context and the metric used to judge it, the basis for interpreting the study's research questions in Chapter 5.

4.11 Summary of Results

The final analytical sample contained 2,229 firms, of which 331 were exporters; Kenya recorded the highest exporter rate (17.48%) and Uganda the lowest (10.91%). Data-quality assessment identified country-level differences in missingness, greatest in Tanzania, and structural, reconciliation and modelling-sample validation were completed successfully for all three countries. Across the five models evaluated, BART-MIA produced the strongest test ROC-AUC and PR-AUC in Kenya and Tanzania, while Random Forest led ROC-AUC in Uganda and the logistic models led PR-AUC there. Confusion matrices showed non-exporters were classified more successfully than exporters throughout, reflecting class imbalance, and Random Forest's cross-validation performance exceeded its test-set performance in all three countries, most sharply in Tanzania and Uganda. Overall, no single model dominates across all countries and criteria — a finding that provides the foundation for Chapter 5's interpretation of these patterns in relation to the research questions, the literature, the Micocci and Rungi (2023) benchmark, and their implications for SME export participation.

CHAPTER 5. DISCUSSION

5.1 Introduction

This chapter discusses the empirical findings from Chapter 4 in relation to the research questions, theoretical framework and empirical literature. The study examined whether observable firm-level characteristics can predict SME exporter status in Kenya, Tanzania and Uganda using five classification approaches: Logistic Regression, LASSO Logistic Regression, CART, Random Forest and BART-MIA.

The study was motivated by the limited evidence on whether machine-learning methods can predict SME export participation in an East African context. The literature review established that exporters and non-exporters differ systematically in resources, productivity, size, financial capacity and technological capability, while heterogeneous-firm theory explains export participation partly through firms' differing ability to absorb the fixed costs of entering foreign markets, and the Resource-Based View (RBV) provides a complementary perspective emphasising differences in the resources and capabilities firms control.

The empirical findings show that exporter status contains sufficient structure to be predicted from observable firm characteristics, but also that predictive performance varies across countries, algorithms and evaluation criteria. BART-MIA produced strong discrimination in Kenya and Tanzania, while Random Forest performed particularly strongly in Uganda; at the same time, high overall accuracy did not consistently translate into high recall for exporters, underscoring the importance of accounting for the substantial imbalance between exporting and non-exporting firms.

The discussion is organised around the principal findings and their relationship to the research questions. It first considers what the predictive results imply for the determinants and structure of SME export participation, then evaluates the findings against the Micocci and Rungi (2023) benchmark, before discussing practical and theoretical implications. The chapter concludes with limitations and directions for future research.

5.2 Discussion of the Main Findings

5.2.1 Research Question 1: Predictive Features

RQ1: Which firm-level features are most predictive of export status among SMEs in the East African Community?

The results show that firm-level information carries useful predictive value: across Kenya, Tanzania and Uganda, the models achieved meaningful out-of-sample discrimination, indicating that exporters and non-exporters can be distinguished to a measurable extent using the selected firm characteristics. This is consistent with the theoretical framework, the Resource-Based View proposes that differences in organisational resources and capabilities contribute to differences in competitive outcomes, while heterogeneous-firm theory suggests firms differ in their ability to absorb the fixed costs of exporting.

However, the results should not be read as showing that a particular predictor causes export participation. The models combine predictors and identify patterns that improve prediction, so feature importance should be interpreted as predictive contribution rather than causal influence. The study's SHAP component is designed for exactly this purpose, examining which characteristics contribute most strongly to individual predictions, the research explicitly incorporates SHAP to examine not only whether exporter status can be predicted but which observable characteristics contribute to those predictions. The country-specific modelling results further suggest that the importance of firm characteristics is not identical across the three economies, consistent with a conceptual framework that recognises resources and capabilities operating within different institutional and economic environments.

Answer to RQ1: Combinations of observable firm-level characteristics contain useful information for predicting exporter status, but their predictive contribution should be interpreted within country context and not as evidence of causal effects.

5.2.2 Research Question 2: Benchmark Comparison

RQ2: How does model performance on EAC SME data compare with the benchmark established by Micocci and Rungi (2023) on European firm data?

The study deliberately follows Micocci and Rungi's (2023) five model classes, Logistic Regression, LASSO Logistic Regression, CART, Random Forest and BART-MIA providing a systematic basis for evaluating whether the predictive framework transfers from the original high-income European setting to the EAC. The results offer qualified support for transferability: BART-

MIA achieved the highest test ROC-AUC in Kenya (0.8526) and Tanzania (0.7746), while Random Forest achieved the highest ROC-AUC in Uganda (0.7906), so BART-MIA remained highly competitive but did not universally dominate.

This matters because the study was not designed as an exact replication — the benchmark and EAC applications differ in geographical setting, firm populations, data sources and available predictors, and the methodology's purpose is to test transferability rather than reproduce identical numerical results. The findings suggest a given algorithm's effectiveness is context-dependent: BART-MIA's strong performance in Kenya and Tanzania shows the flexible modelling approach can transfer successfully to an EAC environment, while its weaker relative position in Uganda shows the benchmark's model ranking cannot simply be assumed to hold unchanged across countries.

Answer to RQ2: The benchmark framework transfers reasonably well to the EAC, but predictive performance and model ranking are country-dependent; BART-MIA remains highly competitive but is not universally superior.

5.2.3 Research Question 3: Practical Predictive Application

RQ3: Can a context-adapted version of the model deliver a practically useful accuracy gain over the direct European benchmark when applied to the full-sector EAC sample?

The results offer qualified support. All three countries produced models with meaningful out-of-sample discrimination. Kenya in particular, where BART-MIA achieved ROC-AUC of 0.8526 and PR-AUC of 0.6266 while Tanzania and Uganda also produced ROC-AUC values well above random discrimination across the principal models. However, the results also show why predictive models should not be treated as automatic decision-making systems: the exporter class is relatively small in each country, and several models showed substantially stronger specificity than recall. This was particularly evident for BART-MIA in Tanzania, for example, it achieved ROC-AUC of 0.7746 and specificity of 0.9802 but recall of only 0.0588, meaning it could distinguish firms reasonably well across probability thresholds while identifying relatively few exporters at the selected classification threshold.

The practical implication is that predicted probabilities may be more useful than a simple exporter/non-exporter classification: a trade-promotion organisation could use predicted

probabilities to prioritise firms for further assessment rather than automatically accepting or rejecting firms against a single threshold. This is consistent with a research design that treats exporter prediction as a potential decision-support application rather than a replacement for substantive assessment, positioning machine learning as a mechanism for generating out-of-sample predictions from firm characteristics.

Answer to RQ3: Firm-level information can support useful exporter predictions, particularly for ranking and prioritisation, but predictions should be used as decision-support evidence rather than definitive classifications.

5.2.4 Research Question 4: Regional Consistency

RQ4: Are the predictors of export status consistent across the three EAC economies studied, or do they vary by country?

The findings show both regional consistency and country-level variation. The three countries share the same twelve-predictor framework and five modelling approaches, allowing meaningful comparison, yet model performance differed: BART-MIA produced the highest ROC-AUC in Kenya and Tanzania, while Random Forest did so in Uganda. PR-AUC results reinforce this BART-MIA led in Kenya and Tanzania, while the logistic models led in Uganda. These country-level differences extend beyond overall accuracy to discrimination, precision–recall performance and the balance between identifying exporters and non-exporters, consistent with the literature gap identified in Chapter 2 regarding limited harmonised evidence on whether exporter predictors remain consistent across EAC economies.

The results suggest a common regional analytical framework is feasible, but country-specific modelling remains important: Kenya, Tanzania and Uganda share regional characteristics, yet differences in institutional, financial and business environments may influence the predictive relationships the models capture.

Answer to RQ4: The predictors contain broadly useful information across all three EAC economies, but their predictive relationships and model performance vary by country. The evidence supports regional comparability alongside country-specific modelling.

5.3 Theoretical Implications

The findings contribute to the theoretical framework in two main ways. First, they provide empirical support for the Resource-Based View's proposition that firms are heterogeneous in their resources and capabilities: the ability to distinguish exporters from non-exporters using observable characteristics suggests exporter status is associated with systematic differences across firms, giving RBV a useful theoretical basis for understanding why firm-level information carries predictive value. Second, the findings are consistent with heterogeneous-firm theory. Melitz's framework proposes that firms differ in their ability to absorb the fixed costs of entering foreign markets, and the present results similarly indicate that firms within the same country are not homogeneous with respect to export participation.

The study does not, however, directly test the causal mechanisms proposed by either theory; instead, it demonstrates that observable characteristics can be combined to generate predictions. The theoretical contribution is therefore primarily complementary: machine learning provides an additional empirical mechanism for studying firm heterogeneity rather than replacing established theories. In this sense, the findings extend both RBV and heterogeneous-firm theory into a predictive idiom, showing that the systematic differences these frameworks anticipate are detectable using routinely collected survey data rather than only through carefully specified econometric models. This is a modest but useful extension, since it suggests that the same theoretical logic developed largely from high-income, financial-statement-based evidence retains explanatory relevance when the underlying data and institutional setting change substantially, as they do between France and the EAC.

5.4 Practical and Policy Implications

The findings suggest export-promotion agencies could use predictive models to improve the targeting of SME support, ranking firms by predicted export probability before subjecting them to additional human assessment. Such an approach could help distinguish firms requiring basic capability development from firms that may already possess characteristics associated with export participation. However, predictive scores should not determine eligibility automatically, since a low predicted probability does not mean a firm lacks export potential.

The country differences also suggest policy should avoid assuming one algorithm or classification threshold will work equally well across the EAC; Kenya, Tanzania and Uganda should retain

country-level validation even where a common regional framework is used. Where an agency's priority is broad screening of many firms, a model with strong ROC-AUC and specificity, such as BART-MIA in Kenya and Tanzania, may be appropriate for narrowing a large applicant pool; where the priority is maximising the number of genuine exporters identified, a model with higher recall, such as the logistic specifications in Uganda, may be preferable even at some cost to precision. The findings therefore support a human-in-the-loop approach: machine-learning predictions can improve targeting and prioritisation, while final decisions remain informed by business assessments and policy expertise.

5.5 Limitations

Several limitations should be considered. The study uses cross-sectional WBES data and predicts observed exporter status rather than future export entry or sustained export performance, so the results cannot establish whether a currently non-exporting firm will become an exporter in future. The analysis is predictive rather than causal: a predictor that contributes strongly to model performance cannot automatically be interpreted as a causal determinant of exporting. The exporter class is relatively small in all three countries, creating a class-imbalance problem that explains why some models achieve high accuracy while producing relatively low exporter recall. Missingness differs across countries, particularly for Tanzania, which may influence comparative model performance. Finally, the study covers three EAC economies and should not automatically be generalised to all African countries; the pooled EAC sample of 2,229 firms is also an order of magnitude smaller than the French benchmark dataset, which may limit the stability of some model estimates relative to Micocci and Rungi (2023).

5.6 Recommendations for Future Research

Future research should extend the analysis in four directions.

First, longitudinal data could be used to predict future export entry rather than current exporter status, providing a stronger test of the practical value of exporter-readiness prediction.

Second, future studies could expand country coverage to other EAC economies and comparable African markets, testing whether the predictive patterns identified here are genuinely regional.

Third, future research could combine predictive modelling with causal methods to determine whether characteristics identified as important predictors actually influence export participation.

Finally, larger datasets and additional firm-level variables could improve model stability and allow more detailed analysis of nonlinear relationships and interactions. Taken together, these directions would help establish whether the exporter-prediction approach evaluated here can mature from a benchmark-replication exercise into an operational tool that EAC trade-promotion agencies, financial institutions and development organisations could adopt with confidence.

CHAPTER 6. CONCLUSION AND RECOMMENDATIONS

6.1 Introduction

This chapter concludes the study by bringing together the main findings on SME export-status prediction in Kenya, Tanzania and Uganda. It summarises the contribution of the research, answers the overall research problem, presents recommendations for practice and policy, and identifies directions for future research.

6.2 Conclusion

This study examined whether machine-learning methods can predict SME export status in the East African Community using firm-level characteristics. The analysis compared Logistic Regression, LASSO Logistic Regression, CART, Random Forest and BART-MIA using World Bank Enterprise Survey data from Kenya, Tanzania and Uganda, benchmarked against the exporter-prediction framework of Micocci and Rungi (2023).

The findings demonstrate that export status contains meaningful predictive structure. Across the three countries, the models achieved out-of-sample discrimination well above random classification, confirming that observable firm-level information — covering firm size and age, access to finance, innovation, technology, quality certification, ownership and human capital — can usefully distinguish exporters from non-exporters, even using only the kind of data a professionally administered enterprise survey can collect.

The study also demonstrates that model performance is context-dependent rather than uniform. BART-MIA achieved the strongest ROC-AUC and PR-AUC in Kenya and Tanzania, while Random Forest achieved the strongest ROC-AUC in Uganda, and the logistic specifications delivered the highest recall and PR-AUC there. Performance rankings therefore shifted according to whether ROC-AUC, PR-AUC, recall, precision or accuracy was the criterion under consideration, and no single model can be regarded as universally superior across the three EAC economies.

The comparison with Micocci and Rungi (2023) provides evidence that their exporter-prediction framework can be transferred to an EAC SME environment: the same five model classes produced meaningful discrimination when applied to firm-survey rather than financial-statement data, and

BART-MIA remained highly competitive in two of the three countries, echoing its strong performance in the original French application. At the same time, the results show that the European benchmark should not simply be applied without contextual adaptation, since differences in economic environment, data structure and country characteristics visibly affect predictive performance and model ranking. The study therefore concludes that context-adapted machine learning can provide practically useful information for SME export analysis, but that predictions should complement rather than replace conventional economic and policy assessment.

6.3 Contribution of the Study

The study makes three main contributions.

First, it contributes empirical evidence to the relatively limited literature on machine-learning prediction of SME export participation in the EAC, where prior work has been dominated by explanatory rather than predictive approaches.

Second, it extends the methodology of Micocci and Rungi (2023) from a European, financial-statement-based firm context to three East African economies using harmonised survey data, providing direct evidence on the transferability of exporter-prediction methods across substantially different economic and institutional environments.

Third, the study demonstrates the importance of country-specific validation: although Kenya, Tanzania and Uganda share a regional context and the EAC Customs Union, the results show that model performance and predictive patterns vary meaningfully between them, cautioning against treating the region as a single homogeneous market for the purposes of predictive modelling.

The integration of machine learning with the Resource-Based View and heterogeneous-firm theory also provides a framework for connecting predictive modelling with established explanations of firm heterogeneity and internationalisation, while the use of SHAP-based interpretation links predictive performance to interpretable, policy-relevant evidence rather than leaving the models as opaque classifiers.

6.4 Recommendations

Based on the findings, several recommendations can be made.

6.4.1 For Export-Promotion Agencies

Trade-promotion organisations should consider using predictive analytics as a screening and prioritisation tool. Firms can be ranked according to predicted export probability and subsequently assessed through conventional business and export-readiness procedures, rather than having predictions substitute for that assessment.

6.4.2 For EAC Policymakers

A common regional framework for SME export analytics could be developed, while allowing individual countries to maintain country-specific models and validation procedures. The differences observed across Kenya, Tanzania and Uganda demonstrate why a single model or classification threshold should not automatically be applied to all EAC economies.

6.4.3 For Financial and Business-Support Institutions

Predictive information could complement existing assessments when identifying firms that may require export-related finance, training or market-development support. However, model predictions should not be used as standalone lending or eligibility decisions, given the modest recall achieved by some models for the minority exporter class.

6.4.4 For Model Users

Accuracy should not be treated as the sole measure of model quality. Because exporters constitute a minority class, recall, precision, F1-score, PR-AUC and balanced accuracy should also be considered when selecting models and classification thresholds, and the choice of metric should reflect the intended use of the predictions.

6.5 Future Research

Future research should extend this study in several directions.

First, longitudinal data should be used to determine whether machine-learning models can predict future export entry, rather than only current exporter status.

Second, additional EAC economies and other African countries should be included to establish whether the country-level differences observed in this study persist across a larger regional sample.

Third, future research could combine machine-learning prediction with causal methods to determine whether the characteristics identified as important predictors actually influence export participation.

Finally, larger datasets and additional firm-level characteristics could improve predictive stability and allow more detailed investigation of nonlinear relationships and interactions between firm resources.

6.6 Final Conclusion

Overall, this study concludes that machine learning provides a promising complementary approach to understanding SME export participation in the EAC. Firm-level characteristics contain sufficient information to generate meaningful exporter-status predictions, but predictive performance varies across countries and evaluation criteria, and no single algorithm dominates in every setting.

The findings support the transferability of the Micocci and Rungi (2023) framework while demonstrating the importance of contextual adaptation to Sub-Saharan African, survey-based conditions. The evidence therefore favours a country-sensitive, multi-metric and explainable machine-learning approach rather than a universal model applied without local validation.

The practical value of the approach lies not in replacing economic theory or human judgement, but in providing additional, interpretable evidence that can improve the identification, prioritisation and support of SMEs with genuine potential for international market participation across Kenya, Uganda and Tanzania.

REFERENCES

- Ackah, C. et al. (2024) ‘Gender, financial constraints and the productivity gap between male- and female-owned businesses in Ghana’. [Publication details to be verified].
- Barney, J. (1991) ‘Firm resources and sustained competitive advantage’, *Journal of Management*, 17(1), pp. 99–120.
- Beck, T. and Cull, R. (2014) ‘SME finance in Africa’, *Journal of African Economies*, 23(5), pp. 583–613.
- Ben Abdallah, M. et al. (2025) ‘Enhancing export competitiveness of SMEs in Sfax: a machine learning approach using principal component analysis’, *Human Behavior and Emerging Technologies*, 2025, Article 7280932.
- Bernard, A.B., Jensen, J.B., Redding, S.J. and Schott, P.K. (2007) ‘Firms in international trade’, *Journal of Economic Perspectives*, 21(3), pp. 105–130.
- Breiman, L. (2001) ‘Random forests’, *Machine Learning*, 45(1), pp. 5–32.
- Chawla, N.V., Bowyer, K.W., Hall, L.O. and Kegelmeyer, W.P. (2002) ‘SMOTE: synthetic minority over-sampling technique’, *Journal of Artificial Intelligence Research*, 16, pp. 321–357.
- Chen, T. and Guestrin, C. (2016) ‘XGBoost: a scalable tree boosting system’, in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. New York: ACM, pp. 785–794.
- Chowdhury, M.A.S., Chuanmin, S., Sokolová, M., Akbar, A., Ali, Z., Ali, H. and Alam, M.Z. (2022) ‘Assessing the empirical linkage among access to finance, firm quality, and firm performance: new insight from Bangladeshi SMEs’, *Frontiers in Psychology*, 13, Article 865733.
- East African Community (n.d.) *About EAC*. Available at: East African Community (Accessed: 23 August 2026).
- Freixanet, J. and Federo, R. (2023) ‘Learning by exporting: a system-based review and research agenda’, *International Journal of Management Reviews*. Early view.
- Geda, A. and Yimer, A. (2023) ‘The trade effects of the African Continental Free Trade Area: an empirical analysis’, *The World Economy*. Advance online publication.
- Gerschewski, S., Scott-Kennel, J. and Rose, E.L. (2020) ‘Ready to export? The role of export readiness for superior export performance of small and medium-sized enterprises’, *The World Economy*, 43(5), pp. 1253–1276.

- Gkypali, A., Love, J.H. and Roper, S. (2021) 'Export status and SME productivity: learning-to-export versus learning-by-exporting', *Journal of Business Research*, 128, pp. 486–498.
- Haddoud, M.Y., Beynon, M.J., Jones, P. and Newbery, R. (2018) 'SMEs' export propensity in North Africa: a fuzzy c-means cluster analysis', *Journal of Small Business and Enterprise Development*, 25(5), pp. 769–790.
- Haddoud, M.Y., Onjewu, A.-K.E. and Nowiński, W. (2021) 'The determinants of SMEs' export entry: a systematic review of the literature', *Journal of Business Research*, 125, pp. 262–278.
- Hansen-Addy, A.E., Parrilli, M.D. and Tingbani, I. (2023) 'The impact of trade facilitation on African SMEs' performance', *Small Business Economics*.
- Hersi, A. et al. (2025) 'Economic integration and trade performance: assessing the effectiveness of the East African Community as a trade bloc', *Cogent Economics & Finance*, 13(1), Article 2516712.
- International Trade Centre (2025) *SME Competitiveness Outlook 2025: A Digital Transformation Roadmap*. Geneva: International Trade Centre.
- James, G., Witten, D., Hastie, T. and Tibshirani, R. (2021) *An Introduction to Statistical Learning with Applications in R*. 2nd edn. New York: Springer.
- Kazimoto, D. and Baadel, S. (2025) 'Empowering Africa's disfranchised SMEs: machine learning-based credit scoring for informal African merchants', *Human-Centric Intelligent Systems*, 5(3), pp. 376–384.
- Konte, M. and Tetteh, G.K. (2023) 'Mobile money, traditional financial services and firm productivity in Africa', *Small Business Economics*, 60(2), pp. 745–769.
- Kumar, S.R., Basavaraj, M.S. and Pachava, V. (2025) 'Machine learning-driven export forecasting: a comparative analysis for MSME growth', *Journal of Economics, Innovative Management and Entrepreneurship*, 3(1), pp. 14–24.
- Love, J.H. and Roper, S. (2015) 'SME innovation, exporting and growth: a review of existing evidence', *International Small Business Journal*, 33(1), pp. 28–48.
- Lundberg, S.M. and Lee, S.-I. (2017) 'A unified approach to interpreting model predictions', *Advances in Neural Information Processing Systems*, 30.
- Masenyetse, R. and Manamathela, M. (2023) 'Firm growth, exporting and information communication technology (ICT) in Southern Africa', *Journal of Innovation and Entrepreneurship*, 12(1), Article 8.
- Melitz, M.J. (2003) 'The impact of trade on intra-industry reallocations and aggregate industry productivity', *Econometrica*, 71(6), pp. 1695–1725.

Micocci, M. and Rungi, A. (2023) 'Predicting exporters with machine learning', *World Trade Review*, 22(5), pp. 584–607.

Mmadubuko, M., Gyan-Amponsah, P. and Sousa, C. (2025) 'Product innovation and SME export intensity: the moderating role of government support, business environment and quality certification', *The International Journal of Entrepreneurship and Innovation*, 26(3), pp. 210–223.

Mpofu, O. and Sibindi, A.B. (2022) 'Informal finance: a boon or bane for African SMEs?', *Journal of Risk and Financial Management*, 15(6), Article 270.

Pacheco, L., Lobo, C. and Maldonado, I. (2022) 'Do ISO certifications enhance internationalization? The case of Portuguese industrial SMEs', *Sustainability*, 14(3), Article 1335.

Panda, S. and Sharma, R. (2023) 'Does innovation spur export performance across countries? An investigation from meta-regression analysis', *Millennial Asia*, 14(3), pp. 341–359.

Phan, T.H., Stachuletz, R. and Nguyen, H.T.H. (2022) 'Export decision and credit constraints under institution obstacles', *Sustainability*, 14(9), Article 5638.

Ringo, D.S., Kazungu, I. and Tegambwage, A. (2022) 'The effect of entrepreneurial orientation on export performance: evidence from manufacturing SMEs in Tanzania', *Cogent Business & Management*, 9(1), Article 2157769.

Sparapani, R., Spanbauer, C. and McCulloch, R. (2021) 'Nonparametric machine learning and efficient computation with Bayesian additive regression trees: the BART R package', *Journal of Statistical Software*, 97(1), pp. 1–66.

Teplova, T. et al. (2022) 'The multifaceted sustainable development and export intensity of emerging market firms under financial constraints: the role of ESG and innovative activity', *Complexity*, 2022, Article 3295364.

UNCTAD (2024) *Economic Development in Africa Report 2024: Unlocking Africa's Trade Potential: Boosting Regional Markets and Reducing Risks*. Geneva: United Nations Conference on Trade and Development.

Wellalage, N.H. and Locke, S. (2016) 'Informality and credit constraints: evidence from Sub-Saharan African MSEs', *Applied Economics*, 48(29), pp. 2756–2770.

World Bank (2020) *Small and Medium Enterprises (SMEs) Finance*. Washington, DC: World Bank Group.

World Bank Enterprise Surveys (2023) *Tanzania Country Profile*. Washington, DC: World Bank Group.

World Bank Enterprise Surveys (2025) *Kenya Country Profile*. Washington, DC: World Bank Group.

World Bank Enterprise Surveys (2025) *Uganda Country Profile*. Washington, DC: World Bank Group.

World Bank Group (n.d.) *Enterprise Surveys*. Available at: World Bank Enterprise Surveys (Accessed: 23 August 2026).

WTO (2016) *World Trade Report 2016: Levelling the Trading Field for SMEs*. Geneva: World Trade Organization.

APPENDICES

APPENDIX A: VARIABLE DICTIONARY

Table A.1 sets out the target variable and the twelve predictors used in the modelling analysis (Table 3.1), together with the specific World Bank Enterprise Survey (WBES) variable codes referenced during the missingness assessment in Chapter 4.3, and the harmonisation treatment described in Chapter 3.4.3. Where a specific WBES code is not individually named in the underlying analysis chapters, the table records that the variable was constructed through the cross-country harmonisation procedure described in Section 3.4.3 rather than taken directly from a single, identically labelled question across all three country instruments; the exact source code for such variables was confirmed during data preparation once the coding and response structures of the three country datasets had been directly inspected, consistent with the study's harmonisation-before-pooling principle.

Table A.1: Variable Dictionary

Construct	Role	WBES Code(s)	Operational Measure
Export participation	Target	d2 (+ related sales-destination items)	Binary exporter/non-exporter classification: 1 if direct exports > 0 or indirect exports > 0, else 0
Firm size	Predictor	Confirmed at data preparation	Permanent full-time employees (5-99 employee SME range)
Firm age	Predictor	b7	Years since operations began, from establishment's reported year of commencement
Access to finance	Predictor	b3a, c39, k30	Harmonised credit-line/loan/overdraft and external-financing-use indicator
Innovation	Predictor	Confirmed at data preparation	Comparable product/process innovation measure
R&D	Predictor	Confirmed at data preparation	Comparable R&D activity/expenditure measure, if available
Technology	Predictor	Confirmed at data preparation	Comparable technology-adoption measure (e.g. foreign-licensed technology)
Digital capability	Predictor	Confirmed at data preparation	Website/digital-use indicator, if consistently available
Quality certification	Predictor	Confirmed at data preparation	Internationally recognised certification
Foreign ownership	Predictor	e6 (+ related ownership items)	Comparable foreign-ownership share measure
Human/managerial capability	Predictor	h1	Training/managerial measure, where comparable
Sector	Predictor/control	Confirmed at data preparation	Harmonised sector classification
Country	Predictor/control	n/a	Kenya, Uganda or Tanzania

Harmonisation notes

Export participation (target variable): constructed from the WBES sales-destination battery (variable d2 and related items), which records the percentage of an establishment's sales attributed to domestic markets, direct exports and indirect exports through third parties. A firm is coded as an exporter (1) where direct exports are greater than zero or indirect exports are greater than zero, and as a non-exporter (0) otherwise. This is the variable identified as carrying the largest share of missing observations in Tanzania during the data-quality assessment (Chapter 4.3).

Firm age: derived from the establishment's reported year of commencement (variable b7), used to compute accumulated years of operation at the time of the survey. The b7 field showed a secondary concentration of missingness, particularly in Tanzania.

Ownership characteristics: constructed from the WBES ownership battery (variable e6 and related items), capturing the percentage of the establishment owned by private domestic individuals, private foreign individuals or companies, and other ownership categories; foreign ownership share is derived from this battery.

Access to finance: harmonised from the WBES finance module (variables including b3a, c39 and k30), which cover credit lines, loan or overdraft facilities and the use of external financing for working capital or investment. Where the three country instruments used different codes to capture an equivalent underlying concept, a single harmonised finance indicator was constructed rather than treating the country-specific codes as separate features, in line with the harmonisation principle set out in Section 3.4.3.

Human and managerial capability: harmonised from WBES workforce and training items (including variable h1 and related codes), covering the availability of formal training programmes and comparable managerial-experience indicators, retained only where consistently available across the three survey rounds.

Innovation, R&D, technology and digital capability, quality certification, firm size, sector and country: constructed as described in Section 3.4.2 from the corresponding WBES modules (innovation and technology adoption, quality-certification status, permanent full-time employment, and the harmonised industry and country identifiers). Because the finalisation of

exact source codes for these items was deliberately deferred until the three country datasets were directly inspected (Section 3.4.1), this thesis does not assert single, universally identical WBES codes for them across Kenya, Uganda and Tanzania; instead, each was reconstructed to represent the same underlying concept, consistent with the study's harmonisation-before-pooling principle.